

The Costs of Building Walls: Immigration and the Fiscal Burden of Aging in Europe*

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Abstract

In low-fertility societies, reducing working-age immigration creates a convex policy frontier: it disproportionately raises dependency ratios, requiring higher tax burdens. We quantify this mechanism using a demographic model and novel estimates of immigrants' fiscal contributions in the euro area, where lifetime net tax contributions are negative. Halting immigration increases the average tax burden by 1 percent of GDP, or €845/year for the average 30-year-old native. Immigration improves fiscal balances even when migrants are low-skilled, if their lifetime contributions exceed those of newborn natives. Raising fertility offers no comparable relief. Differences across countries are substantial, reflecting their positions on the frontier.

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JEL Classification: E62; F22; H55; J11

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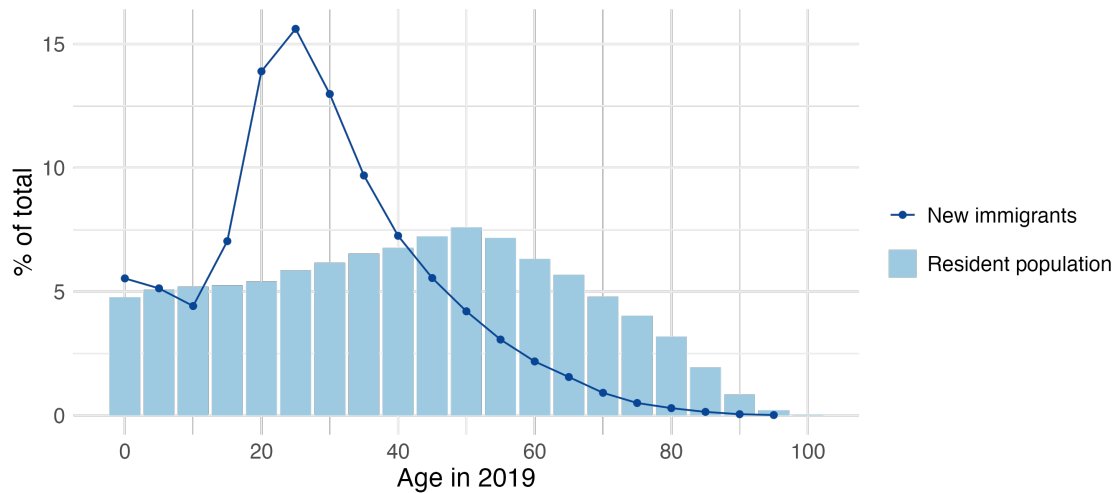
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1 Introduction

Anti-immigration policies are often justified by the view that low-skilled immigrants depress native wages, strain public services, and impose a fiscal burden on native workers. This perspective has shaped policy debates in both Europe and the United States, where restrictions on immigration flows are increasingly popular. At the same time, low fertility rates are causing populations in these countries to age rapidly, while immigrants tend to be young: in Europe, individuals immigrating from outside the European Union (EU) are considerably younger than the resident population, as Figure 1 shows. In this paper, we investigate how these demographic trends shape the fiscal effects of immigration. We find that they imply immigrants can relieve the fiscal burden on native workers even when they are predominantly low-skilled, and that this is currently the case in Europe.

Figure 1: Age Distribution of the Population Stock and the Immigration Flow to the EU in 2019.



Note: Each bar represents the percentage of the resident population in the EU belonging to a specific age group in 2019. The solid line represents the percentage of immigrants arriving in the EU from outside the union in 2019, also segmented by age group. The figure shows that current immigration flows are concentrated among the young working-age population, ages 20–40, rather than the older resident population.

Source: Eurostat.

We study this mechanism of demographic composition by examining how immigration affects public finances in low-fertility societies. We demonstrate a negative, convex relationship between immigration intensity and the fiscal burden. This relationship arises purely from demographic dynamics and does not rely on assumptions about immigrants' productivity, skill composition, or labor market complementarities. We formalize the mechanism and quantify its importance for each euro area (EA) country, using new estimates of immigrants' contributions to government budgets. We find that, due to the age composition of immigration, as long as flows are not interrupted,

extra-EU migrants relieve the fiscal burden of aging by around 1% of GDP per year, on average—despite having lower lifecycle net contributions to the budget than natives.

We begin by examining the properties of the cohort-component model with constant immigration flows, which is the standard framework used by statistical offices for population projections and in the academic literature.¹ We show that, at any finite horizon, restricting immigration leads to higher dependency ratios at an increasing rate. In low-fertility societies, native populations are on a path toward decline to zero, which entails large and growing dependency ratios. Working-age immigration averts this path, sustaining the population and lowering dependency ratios in the long run. At any finite horizon, larger immigration flows raise the share of immigrants and their descendants in the population, shifting the age distribution toward the younger profile of the immigrant group. But this convergence is bounded, giving rise to the convexity: each additional increment to immigration has a smaller marginal effect on dependency ratios, as these approach the transition path of a population composed entirely of migrants.

We build population projections by age, gender, education, and country of birth for each EA country, assuming constant net immigration inflows and varying their levels across scenarios. Immigration inflows are concentrated in the working-age population at the EA aggregate level, albeit with significant heterogeneity across countries. In our baseline, net immigration remains at 0.4 percent of the population per year. Reducing immigration leads to sharply higher dependency ratios, with the increase accelerating as flows are cut further.

We then combine the projections with detailed evidence on immigrants' fiscal contributions across the EA to quantify the fiscal implications. Drawing on multiple European survey sources, we estimate age profiles of taxes and benefits for immigrants and natives separately, conditional on gender and education, using a quasi-saturated regression model. These estimates yield a novel decomposition of the net fiscal contribution for each demographic group in each EA country. By merging them with population projections under varying immigration scenarios, we construct counterfactual government budget paths. We measure the accumulated fiscal imbalances in each, summarizing by the tax increase required to ensure long-run fiscal sustainability without any other change in the government budget. This *rebalancing tax increase* translates population dynamics into a comparable economic measure of the fiscal costs of aging.

¹A constant net migration flow is the standard assumption in population projections. External factors, linked to conditions in countries of origin, are the most important drivers of this inflow and are difficult to predict. All our results hold qualitatively regardless of fluctuations in immigration flows, provided net migration remains positive.

Our headline result is that current immigration flows in the EA reduce the rebalancing tax increase from 16.3% to 14.0%, a 16% reduction. Put differently, shutting down immigration would impose an additional tax burden on resident taxpayers worth 1 percent of GDP per year, or require equivalent spending adjustments. These numbers rely on sustaining the 2019 immigration inflow into the future, both in its overall size and its age-education composition, which skews toward younger, lower-qualified individuals. Further increasing immigration would reduce the fiscal gap, but with diminishing returns: doubling the immigration inflow would lower the rebalancing tax increase by only 1.4 percentage points, a 10% reduction. Because fiscal costs closely track dependency ratios in most countries, a convex policy frontier arises. Restricting immigration raises future dependency ratios at an increasing rate and, consequently, future deficits as summarized by the rebalancing tax increase.

This result does not rely on immigrants being high-skilled. In most countries in our data, immigrants' net fiscal contribution over a full lifecycle is negative and lower than that of natives. Yet because most immigrants arrive as young adults, after childhood and without the associated public expenditures, their remaining lifetime contribution is typically higher than that of a native newborn. In such cases, immigration can improve the fiscal position, even when immigrants are lower-skilled; this fails only when migrants arrive late enough, or benefit systems are generous enough that their net contributions are below those of native newborns. By enlarging the working-age tax base relative to the dependent population, immigration lowers the per-capita adjustment to taxes and transfers that would otherwise be required to restore fiscal balance.

When we look at each EA country separately, we find substantial heterogeneity. The fiscal costs of restricting immigration vary across countries due to differences in baseline immigration levels, the age structure of inflows, and tax-benefit policies. In most countries, including France, Germany, Italy, and Spain, immigrants relieve the fiscal burden of aging, even though their expected net contributions are negative, which drives our aggregate result. In a few cases, however, immigrants' lifetime contributions fall short of those of native-born newborns, and immigration worsens the fiscal position. These cross-country differences imply that a uniform EU migration policy may not be optimal; some countries would benefit from larger immigration inflows than others.

The same mechanism explains why boosting native fertility is not a viable alternative to immigration in averting the fiscal burden of aging. Raising native fertility to replacement reduces the rebalancing tax increase by only 0.7 percentage points, less than one-third of the 2.3 percentage points relief obtained by maintaining 2019 immigration levels. The

difference reflects the timing of fiscal contributions: incoming working-age migrants pay taxes immediately and receive net benefits only later in life, whereas newborns incur spending before entering the workforce. The short-run fiscal costs of a higher share of children thus outweigh the long-run benefits of a larger working-age population, which materialize only decades later.

Our main findings are robust to alternative assumptions regarding the fertility of natives, immigrants, and their offspring, immigrants' educational composition, real productivity growth, and interest rates.

The convexity we highlight arises from demographic dynamics that apply broadly whenever native fertility falls below replacement, and there is a sustained inflow of working-age migrants. This mechanism is therefore not specific to Europe but extends to other low-fertility economies, including Australia, Canada, Japan, Korea, the United Kingdom, and the United States. The United States, in particular, has experienced large inflows of young, predominantly low-qualified immigrant workers broadly comparable in scale and composition to recent flows into the EU. The mechanism we identify further implies that the fiscal gains from expanding immigration would be largest in countries such as Japan and Korea, where immigration flows remain small relative to the resident population and native fertility is particularly low.

Related Literature and Contribution. Our paper relates to three strands of the literature. First, we contribute to the literature on the fiscal effects of immigration in advanced economies. A central finding in this literature is that the fiscal impact of immigration depends critically on its skill composition or indirect effects. Storesletten (2000) shows that immigration can alleviate the fiscal burden of aging in the US, but only if inflows are young and skilled. Applications to European countries yield mixed results: Hansen, Schultz-Nielsen, and Tranæs (2017) find that non-Western immigrants have negative fiscal impacts in Denmark, while Busch et al. (2020) show small positive aggregate welfare effects from Germany's 2015–16 refugee wave, with gains for skilled natives offsetting losses for low-skilled natives. These contrasting findings reflect differences in welfare state generosity, labor market institutions, and calibration assumptions across countries. Other contributions have emphasized indirect general-equilibrium channels: Colas and Sachs (2024) show that complementarity between low-skilled immigrants and high-skilled natives raises native wages and generates indirect fiscal gains, while Clemens (2022) shows that immigration-induced capital accumulation generates additional gains

through capital taxation.²

We contribute to this literature by isolating a demographic mechanism that does not rely on high-skilled immigration or general-equilibrium channels. Immigration improves the fiscal position whenever immigrants' remaining lifetime contributions exceed those of native newborns, a condition that holds even when immigrants are lower-skilled and have lower lifetime contributions than natives. We formalize this mechanism, establish a convex relationship between immigration flows and the fiscal burden of aging, and validate it using harmonized microdata across all EA countries. D'Albis, Boubtane, and Coulibaly (2019), using an OECD panel dataset, provide empirical evidence of a positive relationship between immigration flows and fiscal sustainability, suggesting this occurs through dependency ratios.³

Second, we contribute to the demographic economics literature on the consequences of low fertility. An extensive body of work studies stationary-through-immigration populations: Espenshade, Bouvier, and Arthur (1982) and Schmertmann (1992) showed that constant immigration flows can sustain populations with below-replacement fertility, and a related strand (Weil, 1999, 2023) examines the role of fertility rates for fiscal sustainability. These contributions established that immigration can offset demographic decline, but did not characterize the shape of the relationship between immigration intensity and fiscal outcomes. We prove that dependency ratios are strictly decreasing and convex in immigration intensity at all finite horizons, extending the classical stationary-population results to a setting with direct fiscal implications. We also show that increases in native or immigrant fertility have little impact on fiscal sustainability even under optimistic assumptions, establishing that the immigration margin is quantitatively far more important than the fertility margin for addressing aging.

Finally, our paper relates to the political economy literature on immigration. Several studies show that the fiscal effects of immigration are a first-order driver of anti-immigration sentiment in European countries (Dustmann and Preston, 2007; Hanson, Scheve, and Slaughter, 2007; Alesina, Stantcheva, and Teso, 2018), and Mayda, Peri, and Steingress (2022) document the impact of immigration on voting (see Alesina and Tabellini (2024), for an overview). Our finding that the relationship between immigration and fiscal burden is convex and heterogeneous across EA countries suggests that a uniform EU

²Alongside these works on lifetime contributions, through life-cycle accounting (Auerbach & Oreopoulos, 1999; Lee & Miller, 2000; Razin & Sadka, 1999; Storesletten, 2003), overlapping-generations models (Busch et al., 2020; Storesletten, 2000), or empirical macroeconomic methods (D'Albis, Boubtane, & Coulibaly, 2019; Furlanetto & Robstad, 2019), there is a complementary strand that estimates static contributions at a given point in time (Dustmann & Frattini, 2014; Preston, 2014).

³Two recent policy-oriented papers (Christl et al., 2022; Fiorio et al., 2023) also document short-run positive dividends from immigration in Europe.

migration policy would have asymmetric fiscal consequences across member states.

Outline. The rest of the paper is organized as follows. Section 2 presents the population projections and the theoretical results. Section 3 describes how we estimate the demographic profiles of different budget items and the current net contributions to the government budget of each demographic group. Section 4 quantifies the role of immigration for long-run European public finances. Section 5 shows the cross-country results. Section 6 concludes.

2 Immigration in Low-Fertility Populations: Theoretical Insights and Projections for Europe

In this section, we begin by presenting a three-generation model to characterize how working-age migration affects the age structure of low-fertility populations. We later extend this simple framework into a full cohort-component model, which we use to project the population of each EA country until 2100.

2.1 A Three-Generation Population Model

The population comprises three generations: the young, the working-age, and the old. There are two birth groups: natives and foreigners. The number of persons alive at date t of generation $a \in \{y, w, o\}$ and birth group $i \in \{N, F\}$ is defined as $P_t^{a,i}$. For simplicity, there is a single gender.

There is an exogenous immigration flow, which may vary over time. Immigrants enter the population only at the beginning of their working age. This simple assumption captures the feature of the data that immigration flows from the developing world into advanced countries are mostly composed of younger working-age adults, as shown in Figure 1.

For both natives and immigrants, each period, children achieve working age with probability π_w , adults retire with probability π_o , and the old face mortality risk π_m . Adults have children with a fertility rate f . This Markov chain structure captures the life cycle: time spent in youth before becoming productive and fertile, adulthood followed by retirement, and old age, with a positive probability of death. We assume these parameters are time-invariant. Note that in this model, within each birth-group, replacement fertility is simply given by π_o .

In summary, the population stock of each group evolves over time t according to the following system of difference equations:

$$\begin{aligned} P_t^{O,i} &= (1 - \pi_m)P_{t-1}^{O,i} + \pi_o P_{t-1}^{W,i} \\ P_t^{W,i} &= (1 - \pi_o)P_{t-1}^{W,i} + \pi_w P_{t-1}^{Y,i} + M_t^i \\ P_t^{Y,i} &= (1 - \pi_w)P_{t-1}^{Y,i} + f P_{t-1}^{W,i}, \end{aligned} \tag{1}$$

with $M_t^i = 0$ for $i \neq F$. Variables where i is omitted represent the sum across birth-groups, i.e., $P_t^a = \sum_i P_t^{a,i}$. Throughout, we maintain assumptions A1–A4 from Appendix A; in particular, native fertility is below replacement, $f < \pi_o$, as observed in Europe.

Our analysis focuses on the dependency ratio as the key demographic outcome—i.e., the ratio of non-working (young and old) to working-age population, defined as

$$D_t \equiv \frac{P_t^Y + P_t^O}{P_t^W}.$$

Under a balanced budget, the dependency ratio is a sufficient statistic for the net tax burden. This holds as long as tax and benefit systems maintain the feature that taxes are mainly paid by working-age taxpayers while social benefits and government-provided services (such as education and healthcare) are disproportionately enjoyed by children and the elderly. As the dependency ratio rises, maintaining fiscal balance requires either raising tax rates on the working-age population, reducing benefits and spending for dependents, or some combination of both.

With this in mind, we study the properties of the dependency ratio across different migration-flow paths. First, we establish some properties regarding population stationarity, following standard results in demography. All proofs are in Appendix A.

Lemma 1. (*Extinction path*) Under zero migration ($M_s = 0 \forall s$), the population shrinks to extinction. Along the extinction path, the dependency ratio converges, $D_t^{\text{ext}} \rightarrow_t D_\infty^{\text{ext}}$.

When the native fertility is below the replacement rate, and there are no migration flows, population growth is negative. In the limit, the total population is zero, i.e., it becomes extinct. However, Lemma 1 points out that, along the path to extinction, the dependency ratio converges to a well-defined level.⁴

Population extinction can be avoided with a persistent positive net migration flow.

⁴Populations are modeled as continuous stocks, a standard approach in demography. While this approximation would break down for very small populations due to indivisibilities at the integer level, it is well-suited to the population scales considered here, which are typically in the millions.

Lemma 2 describes the key properties of such a population, known as stationary-through-immigration (SI) in demography.

Lemma 2. (*SI population*) A constant positive inflow of working-age migrants ($\forall s : M_s = M > 0$) allows the population to converge to a stationary-through-immigration (SI) state (Espenshade, Bouvier, & Arthur, 1982). The limiting dependency ratio D_∞^{SI} is independent of the scale of migration flows M .

A constant positive flow of immigrants prevents population extinction even if fertility rates are below replacement for both natives and immigrants. This leads to a stationary population, where population growth is zero in the limit, and the age structure is stable.⁵ The SI population consists exclusively of immigrants and their descendants, as the native group is still in the process of becoming extinct. We now consider the behavior of the dependency ratio along the transition to an SI state.

Proposition 1. (*Effect of migration on dependency ratios*) Consider an initial population P_0 that receives constant working-age inflows $M \geq 0$, with the associated total population dependency ratio at horizon t denoted by $D_t(M)$. For any finite horizon $t \geq 1$, $D_t(M)$ is strictly decreasing and strictly convex in M .

At any finite horizon, the dependency ratio is strictly decreasing in M . Intuitively, new migrant inflows add working-age cohorts, while the incumbent population is a mixture of older cohorts whose dependency ratios have risen over time, as fertility creates young dependents and retirement creates old dependents faster than the working-age stock is replenished (because fertility is below replacement). Increasing the size of the inflow, therefore, raises the worker share at every finite horizon, although with diminishing effect: each additional unit of migration lowers the dependency ratio by less than the previous one, which delivers the strict convexity.

In reality, immigration inflows cannot persist indefinitely, e.g., due to changes in the conditions in origin countries or congestion in the host country. As it turns out, the key results above also apply when immigration flows are interrupted at some future date.

Corollary 1. (*Temporary migration*) Suppose inflows are constant at $M > 0$ until date T^* , and zero thereafter. Then:

1. For all $t < T^*$, the results of Proposition 1 apply.

⁵Note that this is the case even with fertility above replacement for immigrants. Espenshade, Bouvier, and Arthur (1982) show that it suffices for immigrants' descendants to have fertility below replacement from some n -th generation onward to obtain zero population growth in the limit.

2. For all $t \geq T^*$, the population reverts to an extinction path as per Lemma 1.

After T^* , without new migrants, the population still goes extinct in the long run, and the dependency ratio converges to its limiting extinction-path level; however, at any finite horizon before extinction, dependency ratios are still lower than in a no-migrants scenario. The same conclusions hold if inflows decay smoothly to zero.

These results establish that in any aging population with below-replacement fertility, migration has positive but diminishing effects on dependency ratios. The tight link between dependency ratios and fiscal balances creates a convex policy frontier: reducing migration entails increasing the net fiscal burdens borne by resident taxpayers, i.e., mainly natives and pre-existing migrants.

All of this analysis hinges on the key assumption that native fertility remains below replacement. Many developed countries, including European countries and the United States, have experienced below-replacement fertility for several decades, and several middle-income countries are following suit. This is not a temporary phenomenon, and the data provide no indication of a reversal of these trends in the near future (Fernández-Villaverde, 2025; Weil, 2023). Several reasons have been pointed out in the literature for this, such as changes in contraceptive use, delays in childbearing, or shifting of preferences/priorities. See Kearney, Levine, and Pardue (2022) for an overview.

Note that these dynamics are present in any population model with this structure. Therefore, the results of most prior papers on the fiscal effects of immigration that use cohort component models also depend on this effect. These dynamics have not been explained in the literature to the best of our knowledge, but should be taken into account when interpreting such results. In Appendix A, we complement this analysis with numerical simulations to illustrate these dynamics.

2.2 A Full Demographic Projection Model

We now extend the model above to a more realistic setting using a cohort-component model that accounts for gender, year-of-birth generations, and time-varying fertility and mortality rates.⁶ The goal is to quantify the relevance of the effects shown above for the link between migration and dependency ratios—and therefore public finances—for countries today, focusing on the case of Europe.

⁶This class of models is used by different statistical offices worldwide, such as the Census Bureau or Eurostat, in their official population projections, meaning the insights here apply generally to population projections used by policymakers around the world.

The population ages range from 0 to 100; $a = 100$ also includes individuals aged above 100. We consider two genders, male and female, and two country-of-birth groups: “natives”, people born in the EU and their descendants, and “migrants”, people born outside the EU and their descendants (in the data, mostly from developing countries). The number of individuals alive in period t with age a , gender g and country of birth group c , given by $P_{t,a,g}^c$, evolves according to

$$P_{t+1,a+1,g}^c = (1 - m_{t,a,g,c})P_{t,a,g}^c + M_{t,a,g}^c, \quad (2)$$

where $m_{t,a,g,c}$ is the mortality rate and $M_{t,a,g,c}$ is the net migration of age a , gender g and country of birth group c . Newborns are given by

$$P_{t,0,g}^c = \sum_{a,g} P_{t-1,a,g}^c f_{t-1,a,g}^c \lambda + M_{t,0,g}^c, \quad (3)$$

where $f_{t,a,g}^c$ corresponds to the fertility rate at time t of the population with characteristics a, g and c . λ is the gender breakdown of newborns, assumed constant.

2.3 Data Sources and Assumptions

We project the population for each EA country, using 2019 as the starting year.⁷ This avoids the effects of the pandemic, which temporarily increased mortality among the elderly. We run our population projection model until 2100. After that year, we impose population size and distribution to be constant in all dimensions.⁸

Fertility rates differ by age and country of birth group and vary over time. We take values for each EA country from the EUROPOP2019 central projection forecasts (Eurostat, 2020). The total fertility rate of natives starts off slightly over 1.4 and slowly increases over time, though still remaining well below the replacement rate of 2.1 children per woman, reaching around 1.6 by 2100 (see Appendix B). First-generation migrants born outside the EU have fertility rates above replacement, according to Eurostat data for 2019, with a total fertility rate of 2.4. We assume this converges to 2.1 by 2100 and that their descendants exhibit the same fertility as natives (our results are robust to these assumptions, as shown in Appendix D).

Mortality rates are also taken from Eurostat data and their EUROPOP2019 assumptions. Life expectancy at birth was around 81 years in 2019 and is projected to rise (and

⁷We do not include Malta and Cyprus in our exercise as they have a large non-resident population in the workforce that does not appear in the migration numbers but is relevant for public finance purposes.

⁸Interrupting the population transition at a later year does not affect our results.

converge across countries) to around 88 years by 2100. Mortality rates differ only by age and gender.⁹

After projecting the population by age, gender, and country of birth, each cohort is further split by education level. We consider three categories of education: primary, secondary, and tertiary.¹⁰ The population is split by education level, also using data from Eurostat that includes differences by country of birth. This is crucial for an accurate measurement of the immigrants' contribution to the budget. As for the future path of education levels, we base it on a conservative invariance assumption. Specifically, we assume that, when all new cohorts reach age 25, the cross-sectional education distribution is the same as that of the 25-year-old cohort in 2019. This implies that the overall education level of the projected population will increase over time as older cohorts, who generally have lower education, are replaced by younger, more educated ones.

As in the three-generations model, the net migration flow does not change over time. This assumption is consistent with official demographic projections, such as those routinely performed by Eurostat or the Census Bureau. These projections account for ongoing shocks that affect net migration inflows, but in the medium and long run, they assume a constant net migration flow. Forecasting future migration shocks is challenging as they depend, for example, on wars or natural disasters that are generally unpredictable.

Later, in our mechanism quantification, we will vary the net migration flow of those born outside the EU, keeping the net migration flow of those born in the EU constant. We also keep the age, gender, and education composition of the flow constant. Note that in all scenarios, total net migration is positive, although it may be negative for some demographic groups. Appendix B describes more details about these population projections.

2.4 Population Projection Results

In 2019, the starting point of our projection, about 342 million people lived in the EA. Table 1 compares the population composition by country of birth in terms of age and education in 2019. Panel (a) of the Table shows the age composition. The non-EU-born population is heavily concentrated in the working-age group, with more than two-thirds

⁹Data on mortality rates by education level are only available for a few EA countries. We could expect projected dependency ratios to be slightly higher if we incorporated such differences, as individuals with higher education levels tend to live longer.

¹⁰Primary education corresponds to ISCED 2 or below in the International Standard Classification of Education (ISCED), secondary corresponds to ISCED 3 and 4, and tertiary corresponds to ISCED 5 and above.

aged between 26 and 65 years. By contrast, the EU-born population is almost evenly split between dependent groups (young and old) and the working-age group.

Table 1: Composition of Population by Country of Birth in 2019.

<i>(a) Age (full population)</i>		
Age in 2019	2019 Population stock	
	EU-born	non-EU-born
0–25 years old	27.7%	17.5%
26–65 years old	52.0%	71.1%
>65 years old	20.3%	11.5%

<i>(b) Education Level (population above 25 y.o.)</i>		
Education attained	EU-born	non-EU-born
Primary	28.8%	41.1%
Secondary	42.3%	33.8%
Tertiary	28.9%	25.1%

Note: The tables above describe the composition of the resident population in the EA in 2019 by age and education, looking separately at groups based on country of birth. Panel (a) shows the age composition of each group, while panel (b) shows the composition in terms of attained education, considering only residents above 25 years old. Primary education corresponds to below high school attainment (ISCED 0–2), secondary education corresponds to completed high school education (ISCED 3–4), and tertiary education corresponds to a university degree (ISCED 5–8).

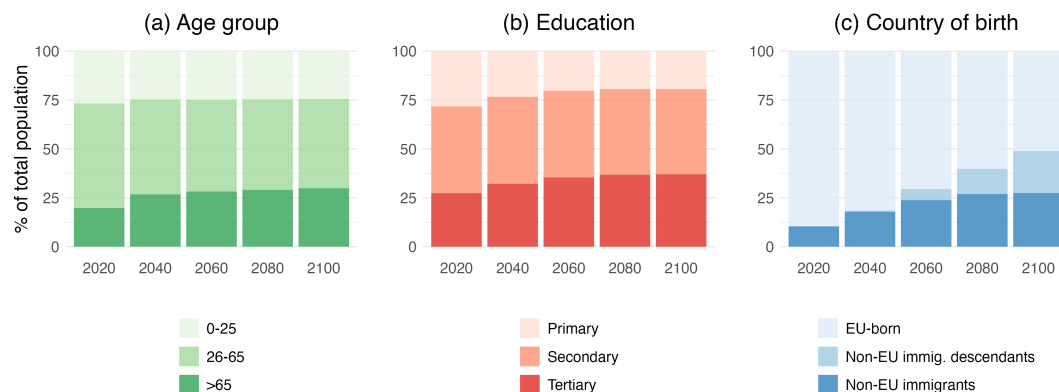
Source: Eurostat.

Panel (b) of Table 1 shows the education composition of the population above 25 years old by country of birth in 2019. While the largest education group for the EU-born population is secondary, for the non-EU-born population it is primary, indicating that the native population is more educated than the non-EU-born population.

Using the population projection model described by Equations (2) and (3), we forecast, for each country of the EA, the population stock until 2100. After that year, we assume a constant population. By 2100, we project that 330 million people will live in the EA. Figure 2 plots, for selected years of the projection, the age, gender, education level, and country of birth distribution for the EA.

In our baseline projection, the population ages significantly. The dependency ratio (the ratio of the young and old populations to the working-age population) increases from 0.89 in 2020 to 1.13 by 2100, in line with Eurostat projections. Panel (a) shows that the share of the working-age population decreases, while the share of the old population increases. The share of the young population first declines due to low fertility rates, then slowly returns to its initial level as fertility rises. At the same time, as life expectancy continues to rise, the share of the population aged 65 or older increases by more than 10 percentage points over the projection time span. The gender distribution, not plotted, is projected to remain constant, with females making up just over half of the total population, as female life expectancy is slightly higher than male life expectancy.

Figure 2: Demographic Distributions Implied by the Population Projections.



Note: The figure shows the percentage of people with given demographic characteristics for selected years of the projection. Panel (a) displays the age distribution by age group (young, working-age, and old-age). Panel (b) presents the education level distribution as a percentage of the adult population (primary corresponds to below high school attainment, secondary corresponds to completed high school education, tertiary corresponds to a university degree). Panel (c) depicts the distribution of country of birth as a percentage of the adult population. The group of non-EU immigrant descendants includes only those born from 2019 onward, with preexisting descendants included in the EU-born group. Finally, the gender distribution is not plotted, but it remains constant around 50% throughout the projection.

Source: Authors' calculations.

Our demographic forecasts also imply that the distribution of education levels of the adult population will change, as panel (b) illustrates. As described above, we assume that the education distribution verified at age 25 in 2019 is the distribution that new cohorts attain once they reach the same age. This implies that the percentage of the population attaining secondary and, especially, tertiary education will increase over time. In fact, the share of the adult population with the highest education level is expected to increase by 12 percentage points, whereas the share with the lowest education level is expected to fall by about 10 percentage points between 2020 and 2100.

Last, panel (c) shows that the share of non-EU immigrants is projected to rise by 15 percentage points, and the children of this group are projected to represent 25 percent of the adult population in 2100. The non-EU-born population has a higher fertility rate than the native population, and our projections assume a constant annual flow of immigrants. At the same time, native fertility is below replacement, meaning the native population is shrinking. Consequently, the population projections imply that the share of non-EU born and their descendants increases over time, representing around 50% of the population by 2100.¹¹

¹¹Our data does not allow us to break down the resident population in 2019 by the country of birth of parents. Eurostat estimates, based on Labor Force Survey data, place the share of the working-age population that is a descendant of at least one foreign-born person (including both intra-EU and extra-EU) at 7.3% for the whole EU in 2023.

3 Fiscal Burden of Aging in Europe and the Contributions of Natives and Migrants

In this section, we describe how we map the different budget aggregates to the demographic groups considered by constructing the government budget's demographic profile for each EA country. We then combine this profile with the population projections to measure the fiscal burden of aging. Our overall methodology is based on the generational accounts of Auerbach, Gokhale, and Kotlikoff (1991), who propose them as an alternative to standard public accounts for measuring the effects of demography on public finances.

3.1 Fiscal Contributions of Different Demographic Groups and the Rebalancing Tax Increase

We divide the population into demographic groups by age, gender, education level, and country of birth, the same groups that we consider in the population projection model. For each of these groups, we want to estimate its net contribution to the government budget. For that, we need to compute how much they pay for each tax item (e.g., personal income tax) and how much they receive and benefit from each expenditure item (e.g., healthcare). Let $\tau_{t,x}^i$ be the average per capita government revenue of category i that is attributed to an individual of demographic group x in the year t , and $g_{t,x}^i$ is the average per capita government expenditure of category i attributed to a person of demographic group x in year t .

The sets $\{\tau_{t,x}^i\}$ and $\{g_{t,x}^i\}$ allow us to build the demographic profile of the government budget in year t . We estimate these profiles for the base year $\bar{t} = 2019$ and assume that they grow at the rate of labor productivity, γ , plus the inflation rate, π .¹²

$$\tau_{t,x}^i = \tau_{\bar{t},x}^i \prod_{j=\bar{t}+1}^t (1 + \gamma_j) (1 + \pi_j), \text{ for } t > \bar{t}, \quad (4)$$

and similarly for government expenditure items. We next derive a metric for the fiscal burden of aging. It is important to note that we do not aim to accurately forecast the different budget aggregates, whose future paths certainly depend on many factors beyond demographics, including policy changes. Our goal is to quantify the fiscal imbalances induced by aging trends under different demographic scenarios.

¹²2019 is a good year to estimate the demographic profiles as it does not suffer from any contamination from the 2020-2021 health crisis and subsequent 2022-2023 cost-of-living crisis.

Following the GA literature, we define public finances to be balanced if the present discounted value of current and future revenues is equal to the present discounted value of current and future expenditures plus current debt, that is, if the intertemporal government budget constraint (IGBC) holds:¹³

$$\sum_{s=0}^{\infty} \prod_{j=1}^s \frac{T_{\bar{t}+s}}{(1+i_{\bar{t}+j})} = B_{\bar{t}-1} + \sum_{s=0}^{\infty} \prod_{j=1}^s \frac{G_{\bar{t}+s}}{(1+i_{\bar{t}+j})}. \quad (5)$$

Total government revenues are the sum of all revenue items, and each revenue item is the sum across demographic groups of the product between the population stock and the demographic profile of that group, i.e., $T_t = \sum_i \sum_x \tau_{t,x}^i P_{t,x}$, and similarly for government expenditures. We introduce a wedge $(1 + \theta_\tau)$ in the above equation, which corresponds to a proportional adjustment factor to revenues necessary for the IGBC to hold. Given a set of demographic profiles at $\bar{t}$, future paths for the population of each group x , and a starting value for public debt $B_{\bar{t}-1}$, this adjustment factor is implicitly given by:

$$\sum_{s=0}^{\infty} \sum_i \sum_{x \in X} \mathcal{D}_t^s \left[g_{t,x}^i - (1 + \theta_\tau) \tau_{t,x}^i \right] P_{\bar{t}+s,x} + B_{\bar{t}-1} = 0, \quad (6)$$

with $\mathcal{D}_t \equiv \frac{(1+\gamma_t)(1+\pi_t)}{1+i_t}$, which represents the growth/discount factor.¹⁴ The adjustment factor θ_τ represents the permanent tax increase, across all revenue categories and groups, necessary to ensure intertemporal fiscal balance, keeping the same demographic structure of the budget. This means this metric does not account for future budget reforms (such as the minimum retirement age), since it assumes demographic profiles remain unchanged. We refer to θ_τ as the *rebalancing tax increase*. Note that this adjustment factor, contrary to the traditional metrics in the generational accounts literature, is robust to assumptions about real productivity growth and interest rates. In Appendix D, we show that.

θ_τ can be seen as the minimum tax increase necessary to balance public finances. Given fixed demographic profiles, delaying adjustment means increasing revenues only later, thereby widening the tax increase that would then be required in future periods. In practice, the adjustment implied by θ_τ might lead to an impractical public debt trajectory, such as sustaining large surpluses for an extended period. In this case, the government would accumulate assets that would later be drawn down to sustain small primary deficits. Note also that the focus on tax increases is simply for interpretability, as the

¹³To be precise, the IGBC should also include the term $\lim_{s \rightarrow \infty} \prod_{j=1}^s \frac{B_{\bar{t}+s}}{(1+i_{\bar{t}+j})}$. Public finance sustainability implies that the government is solvent; hence, Ponzi schemes cannot occur.

¹⁴For simplicity, we assume that the productivity growth rate, the inflation rate, and the interest rate are constant and equal to the historical long-run average.

same exercise could be performed by introducing an adjustment factor to government expenditure, which would represent the permanent change required across all expenditure categories to ensure the IGBC holds. The results of doing so are essentially symmetric, and therefore, for presentation purposes, we focus only on the rebalancing tax increase.¹⁵

Since we are estimating the profiles for a given base year and then projecting them, we may also be projecting year-specific effects. Specifically, $\bar{t}$ was associated with a given business cycle position, which affected the budget, such as higher unemployment benefits when $\bar{t}$ was a recession year, or higher income taxes when the economy was expanding that year. To address these effects, we use cyclical adjustments to fiscal aggregates, following the approach of Bonin, Patxot, and Souto (2014), who apply the cyclically neutral budget adjustments of Girouard and André (2006) to a generational accounting exercise. In Appendix B, we describe this adjustment in detail.

Furthermore, our framework assumes prices do not adjust. Because immigration raises employment, holding prices fixed implies a corresponding increase in the capital stock, which in turn generates additional capital income and higher tax revenue. This bias is especially relevant in our setting, as immigrants are disproportionately of working age. To address it, we follow Clemens (2022) and apply their proposed adjustment. While this correction has quantitative implications, it does not qualitatively alter our main conclusions. Appendix B describes the procedure in detail.

Note that θ_τ is positive when the discounted sum of the contributions to the budget is smaller than the benefits paid by the government, meaning that restoring fiscal balance requires a tax increase. As we can see in Equation (6), since the demographic profiles grow uniformly for all demographic groups, the value of θ_τ is larger if the demographic groups for whom $\sum_i g_{t,x}^i > \sum_i \tau_{t,x}^i$ grow more over time, and smaller in the opposite case. As we shall see in the data, young and retired groups are net beneficiaries from the budget, and the working-age group is a net contributor. Therefore, the value of θ_τ follows the evolution of the dependency ratio, and hence the results of section 2 also apply to public finances. Following Proposition 1, for all periods of the projection, the dependency ratio decreases with the net migration flow size, and so does θ_τ .

3.2 Data Sources and Methodology

In this subsection, we explain the data and methodology used to estimate the demographic profiles, $\{\tau_{t,x}^i\}$ and $\{g_{t,x}^i\}$. As for the population projections, we obtain data for

¹⁵Blanchard (1990) argues why an indicator such as θ_τ is the most appropriate measure of long-run fiscal sustainability.

all countries that were part of the EA in 2019, except for Malta and Cyprus.

Data. To estimate the demographic profiles, we use two types of data. First, we use individual micro-level data on individuals' demographic characteristics, their payments for each tax, and their benefits from each social subsidy and public service. Second, to ensure our exercise is consistent with the national accounts, we use the macro-aggregate values for the different budget items.

Individual microdata on taxes and benefits come mostly from the EU Statistics on Income and Living Conditions (EU-SILC) survey. This dataset provides timely, comparable cross-sectional and longitudinal data on income, poverty, social exclusion, and living conditions. For our purposes, we use only the cross-sectional dimension of the data set. We use individual data on labor income, income and property taxes, and social transfers, including pensions. This allows us to estimate the demographic profile of personal income tax, social security contributions, property tax, old-age pension, survival pension, disability pension, unemployment benefits, and sickness allowance.

To achieve a more complete coverage of the government budget, we use two additional microdata sources. We obtain household-level consumption data from the Household Budget Survey (HBS) to derive the demographic profile of consumption, which we use to allocate value-added tax to each group x . We also use the European Central Bank's Household Finance and Consumption Survey (HFCS) data on household business wealth holdings, which we take as a proxy measure for the incidence of corporate income tax.¹⁶

For healthcare spending, we obtain an age-gender profile of government health expenditure reported directly by the European Ageing Working Group (European Commission, 2018). For education expenditure, we use Eurostat data reporting expenditures by level of study.¹⁷

Macro-aggregates on taxes and benefits come from Eurostat. We also use data on the productivity growth rate, the inflation rate, the interest rate, and public debt to compute the discount factor. Appendix C contains the list of variables used from the different data sources.

¹⁶Corporate income tax is paid by firms on their profits. However, firms ultimately belong to households; hence, we use the distribution of households' business wealth holdings to measure the incidence of this tax on individuals.

¹⁷We allocate primary education (ISCED 0-1) expenditures to ages 2 to 11, secondary (ISCED 2-4) to ages 12 to 18, and tertiary or higher education (ISCED 5-7) to the population aged between 19 and 25 who completed at least secondary education. This represents a typical situation in European countries, as described by Motiejunaite-Schulmeister, Sicurella, and Birch (2022).

Methodology. To estimate the demographic profiles, we proceed in two steps. First, we estimate a quasi-saturated regression model on categorical variables of country and demographic groups for each tax/benefit item:

$$\begin{aligned}
\tau_{i,j} = & \alpha + \beta^{\text{country}} \cdot \text{country}_j \\
& + \gamma^{\text{age} \times \text{country}} \cdot (\text{age}_j \times \text{country}_j) + \phi^{\text{gender} \times \text{country}} \cdot (\text{gender}_j \times \text{country}_j) \\
& + \delta^{\text{educ} \times \text{country}} \cdot (\text{educ}_j \times \text{country}_j) + \kappa^{\text{CoB} \times \text{country}} \cdot (\text{CoB}_j \times \text{country}_j) \\
& + \eta^{\text{gender} \times \text{educ} \times \text{country}} \cdot (\text{gender}_j \times \text{educ}_j \times \text{country}_j) \\
& + \psi^{\text{age} \times \text{gender} \times \text{country}} \cdot (\text{age}_j \times \text{gender}_j \times \text{country}_j) \\
& + \theta^{\text{educ} \times \text{CoB} \times \text{country}} \cdot (\text{educ}_j \times \text{CoB}_j \times \text{country}_j) \\
& + \lambda^{\text{CoB} \times \text{age}} \cdot (\text{CoB}_j \times \text{age}_j) + \mu^{\text{CoB} \times \text{gender}} \cdot (\text{CoB}_j \times \text{gender}_j) + \varepsilon_j,
\end{aligned} \tag{7}$$

where $\tau_{i,j}$ denotes the value paid or received by individual j of a given budget item, country_j is the vector of the EA country dummies, age_j indicates the individual's age bracket (groups of 5 ages), educ_j denotes the education level group (primary, secondary and tertiary), gender_j is a gender indicator (male and female), and CoB_j is the individual's country of birth group (EU and non-EU). Using this specification, we estimate country-specific profiles for age, gender, education, and country of birth. The interaction terms allow for country-specific gender-education differences, gender-age differences, and education-country-of-birth differences. Furthermore, our specification allows for systematic country-of-birth gaps along the age and gender dimensions. Note that the $\text{CoB} \times \text{age}$ and $\text{CoB} \times \text{gender}$ terms are not interacted with country, so identification of these gaps relies on EA-wide variation, with the gap shape assumed to be common across countries. Nevertheless, country-specific age and gender profiles, and the average country-of-birth effect by country, remain free to differ. We adopt this restriction because the survey data do not support identifying country-specific immigrant-native gaps with reasonable precision; pooling across the EA is necessary for identification with reasonable precision.

In a second step, using the estimated coefficients, we predict for each country, age bracket, gender, skill level, and country of birth, how much, on average, an individual contributes to each revenue item and receives from each benefit item. For items estimated using the HFCS data, we do not distinguish by gender because the unit of observation is the household.

In total, we estimate the demographic profiles for 12 budget items that account for about two-thirds of the budget. The remainder of the budget cannot be mapped to a demographic group, due to data limitations or conceptual reasons. Benefits from government activities such as defense, justice, or regulation cannot easily be distinguished between

individuals. For the revenues of these budget items, we distribute them uniformly to all individuals older than 18 years, and for the expenditures, we distribute them uniformly, independent of any demographic characteristics.¹⁸ Appendix B has additional estimation details of these profiles, and Appendix E reports the profiles for each budget item and each country of the EA.

3.3 Demographic Profile of Revenues and Expenditures for Natives and Migrants

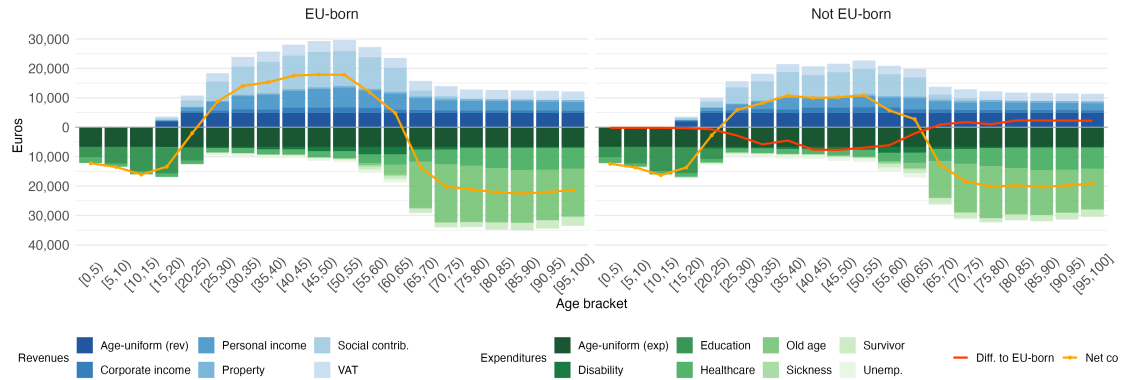
Figure 3 plots the EA mean age profiles of revenues and expenditures per capita, estimated using the microdata sources described above, for individuals born in the EU and outside the EU, respectively, in the left- and right-hand charts. The top blue bars show, for each age group, the average 2019 payment an individual made to the government, broken down by revenue category. Similarly, the bottom green bars show, for each age, the average amount received from the government by expenditure category. The yellow line shows the difference between total revenues and total expenditures for each age and country-of-birth group. This corresponds to the per capita net contribution. The red line in the right panel gives the difference between migrants' and natives' net contributions by age.

For both natives and migrants, the figure shows the expected life-cycle pattern, with three distinct age groups. The first are the younger age groups (0–24 years old), who are net recipients of the budget, benefiting from education and healthcare while paying very little in taxes. Per capita spending on education and healthcare is assumed to be the same for migrants and natives, and estimated revenue differences are small; thus, in this age group, there is essentially no difference between migrants and natives.

The second group consists of individuals in the working-age group (25–64 years old). These groups are net contributors to the budget as they pay more in taxes (personal income tax, VAT, and social contributions are the largest items) than what they receive in benefits and public services. Differences between migrants and natives are the largest here, as working-age migrants provide substantially less revenue in per capita terms, around 5,000 Euros less on average, whereas spending is similar. Most of the difference comes from social contributions and personal income taxes, reflecting migrants' lower average education levels and wages.

¹⁸This distinction is mostly for presentation purposes and does not materially affect the results. In any case, the rationale is that children, while unable to earn any income and thus not bearing the burden of taxation until they reach working age, can and do benefit from public services since birth.

Figure 3: Mean Age Profile of Revenues and Expenditures Per Capita.



Note: The figure shows estimates of mean per capita amounts for the different revenue and expenditure components of the government budget, for the EA average (weighted by GDP). The orange solid line shows the net contribution by age, which is the sum of revenues minus the sum of expenditures. In Appendix E we show the same plot but for each country. *Source: Authors' calculations.*

The third segment of the population is the older age group (65+ years), which is a net recipient of government budget funds. Although they still provide some revenue in income and consumption taxes, they receive substantially more, mostly in the form of pensions and healthcare services. The average migrant receives significantly less than natives at these ages because their pensions are lower, and while they also contribute less in revenue, the difference is much smaller. As a result, their net contribution is more than 2,000 euros higher than that of natives.

These tax and benefit profiles result from combining groups with different education levels among both natives and migrants. Individuals with lower education (primary education attainment) have lower income, which implies they pay fewer taxes during their working ages, but receive benefits at a similar level to the highly educated. On the other hand, more educated individuals (with secondary or tertiary education attainment), who tend to earn higher incomes, tend to pay more taxes and are net contributors to the budget. In Appendix E, we show these mean age profiles by education group.

In Table 2 we show the net contribution by country of birth (EU-born and non-EU-born) and age group (0–25, 26–65, and >65 years old, the three relevant groups). Column (1) shows the net contribution per capita, which is the average value given by the yellow line in Figure 3 over the age brackets.

Column (2) of the same table contains the population of each group, and column (3) corresponds to the respective total contribution to the budget. The same life-cycle pattern of the per capita net contribution emerges here for both country-of-birth groups. The current total net contribution of the EU-born population is negative, whereas the total

Table 2: 2019 Primary Budget Decomposition.

Country of Birth	Age	(1) Net Contrib. (€, per capita)	(2) Pop. (Million)	(3) Contrib. to the Balance (Billion €)	(% GDP)
EU-born	0–25	-10,635	86.9	-858.4	-7.37
	26–65	13,125	161.0	1,927.3	16.55
	>65	-19,546	63.1	-1,141.1	-9.80
	total	–	311.0	-72.3	-0.62
Non-EU-born	0–25	-8,279	5.3	-40.1	-0.34
	26–65	8,158	21.5	165.0	1.42
	>65	-17,219	3.5	-58.9	-0.51
	total	–	30.3	66.0	0.57
Total		–	341.3	-6.3	-0.05

Note: The table shows, by country of birth and age group, the per capita net contribution in column (1), the population stock of that group in column (2), and the total contribution to the balance in billion Euros and in percentage of the potential GDP in column (3), for the EA.

Source: Authors' calculations.

net contribution of the non-EU-born group is positive, which leads to an essentially balanced budget (a small deficit of 6.3 billion euros, which corresponds to 0.05% of the EA potential GDP). In other words, without the contributions of the non-EU-born population group, the EA budget would be unbalanced.

These results relate to the literature that has computed the so-called “static” net contribution of immigrants to the budgets of governments in advanced countries. In particular, they are consistent with the results of Fiorio et al. (2023), who also examine EU data and find that between 2014 and 2018, the per capita net fiscal contribution of migrants in the EU was substantially higher than that of natives due to their age composition. Before, Mackie and Blau (2017) examined the US, finding a similar life-cycle pattern in net contribution and showing that immigrants, particularly second-generation immigrants, have a positive net fiscal impact once they become independent adults. Dustmann and Frattini (2014) found that in the UK between 1995 and 2011, immigrants from the EU had positive net contributions, whereas non-EU immigrants and natives had negative net contributions.

In the next section, we will introduce population dynamics to study the fiscal burden of aging and the potential of immigration to attenuate it. The exercise is to compute the counterfactual government budget balance implied by the population projections. Essentially, we keep the demographic profiles of the budget (that imply the values in column (1) of Table 2) fixed, only letting them grow at a constant rate, and use the population dynamics produced by the model of section 2 to compute net contributions (what we have in column (3) of Table 2) over time.

4 Immigration and the Fiscal Burden of Aging in Europe

In this section, we study the role of immigration flows in shaping the tax burden as it rises with population aging. We start by computing the lifetime expected net contribution of different demographic groups. Second, we build counterfactuals at different levels of immigration intensity, allowing us to quantify the relationship between immigration and the fiscal burden of aging. Third, we present an exercise that computes the fiscal effects of increasing native fertility, to provide a comparison with the impact of immigration. Finally, we present several robustness checks to assess the validity of our main results.

4.1 Lifetime Net Contributions of Immigrants and Natives

Immigration flows in most EA countries are concentrated between ages 20 and 40, as shown in Figure 1. The lifetime contribution of workers entering at those ages will tend to be well above a full-life-cycle contribution, given the age profile of net contributions, negative at youth and positive during the working ages. We measure this by computing expected lifetime net contribution as follows, for a person aged a_e in 2019:

$$\sum_{a=a_e}^{100} \sum_i D^{a-a_e} \left[\tau_{t,a,x'}^i - g_{t,a,x'}^i \right] \left(\prod_{j=a_e}^{a-1} \left(1 - \pi_{j,x'}^X \right) (1 - m_{j,x'}) \right),$$

where D , τ^i , and g^i are defined as before, π^X is the probability of out-migration, m is the mortality rate, all of which depend on age a and other demographic characteristics, which are constant and denoted by x' . In Table 3, we show these lifetime net contributions for both natives and immigrants from outside the EU, computed as of 2019, in three cases: at birth, for a taxpayer aged 30 in 2019, and the expected value given entry age in 2019. For natives, this is, by definition, the same as the contribution at birth; for immigrants, it is weighted by the importance of each age in the 2019 immigration flow.

Table 3: Expected Lifetime per Capita Net Contributions in 2019.

	Net Contribution		
	At age 0	At age 30	Expected given entry age
Natives	-5,051	2,160	-5,051
Immigrants from outside the EU	-9,757	-2,697	-1,002

Note: Annuity values, in Euros per year. Average across countries weighted by the country's potential GDP in 2019. The table shows, by country of birth group, the average projected lifetime net contribution for a person at birth, at age 30, and at the average entry age.

Source: Authors' calculations.

Both EU-born and non-EU-born individuals have negative lifetime contributions over the full life cycle (from age zero) of €–5,051 and €–9,757 per year, respectively; immigrants'

contributions are worse due to their lower levels of education. At age 30, a native taxpayer has a positive contribution over their remaining life cycle, with taxes and contributions during their working life exceeding the expected net costs in retirement; an immigrant at age 30 still makes a negative contribution over their remaining life cycle, albeit considerably smaller. The gap between natives and immigrants is similar at birth and at age 30: since public expenditures on children are very close across country of birth groups, removing the youth phase from the calculation shifts both groups' contributions by a similar amount.

To assess the contribution of immigrants to the government's intertemporal budget constraint, the relevant quantity is the expected lifetime contribution given age at entry. The average new immigrant has an expected remaining-lifetime contribution of -€1,002 per year, which, although negative, is €4,000 above the native-newborn benchmark. Each arriving migrant adds more to the tax base in present value than newborn natives—the other category of new entrants in the population—even though immigrants are lower-skilled and make smaller per-period net contributions during their working years. Nevertheless, the average expected contribution remains negative given entry ages, showing that immigration can relieve but not, per se, eliminate the long-run fiscal imbalance.

4.2 The Effects of Reducing or Expanding Immigration Flows into Europe

We now use our framework to study how varying intensities of net migration flows affect fiscal balances over time. By combining the demographic profiles estimated in section 3 with the population projections computed in section 2, we obtain counterfactual government budget sequences under different scenarios for immigration flows. Then we can calculate, for each scenario, annual budget balances and the rebalancing tax increase, a measure of long-run fiscal balance.

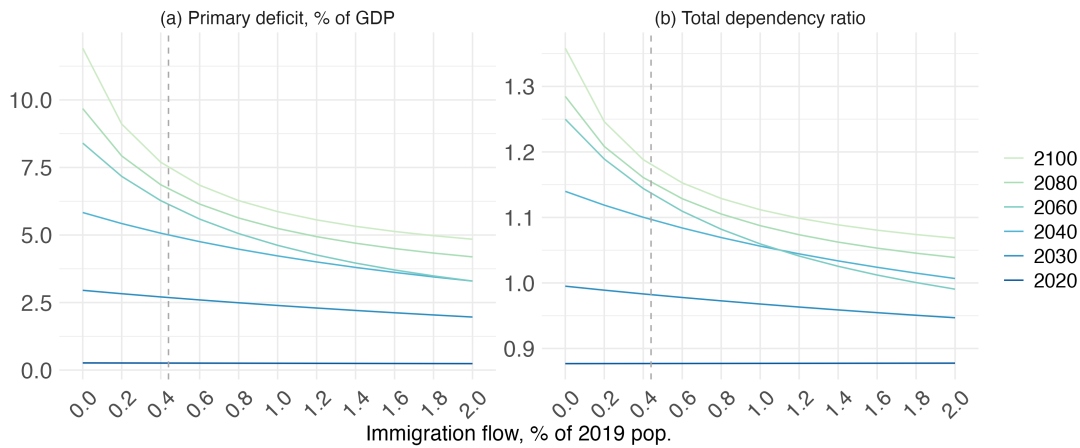
4.2.1 The Effects of Immigration on Future Primary Deficits

Our baseline projection assumes a constant yearly net migration flow from non-EU countries at the level observed in 2019, around 0.4 percent of the EA population in that year. In this baseline scenario, the initial deficit of 0.05 percent of 2019 potential GDP widens over time. By 2040, the primary deficit would be 4.7 percent, and by 2100, it would be almost 9 percent of potential GDP. Appendix E shows the evolution of this variable for the EA average, as well as for individual EA countries, where we generally

observe the same trend. We also show the contribution of each age, gender, education, and country-of-birth group to the primary balance over time.

We focus on the effect on the primary balance of changing the intensity of net migration flows. The age/gender/education distribution of incoming net migration is kept equal to that observed in 2019, with only the scale of net migration flows changing across scenarios.¹⁹ Figure 4 shows the results of this exercise, plotting for selected points in the projection horizon different variables along the migration scenarios. Panel (a) shows the primary deficit in percentage of the potential GDP, and panel (b) shows the total age dependency ratio.

Figure 4: Primary Deficit and Demographic Dynamics under Different Net Migration Scenarios



Note: Panel (a) shows the projected primary deficit as a percentage of the potential GDP for the different net migration scenarios. Panel (b) plots the age-dependency ratio (computed as the sum of the young and old populations divided by the working-age population) for these same net migration scenarios. The dashed line in all three panels indicates the size of migration flows in the baseline scenario.

Source: Authors' calculations.

Panel (a) shows that, regardless of the level of migration, the current small primary deficit will increase over time. These projected deficits are driven primarily by the dependency ratio, as confirmed by panel (b) and consistent with the analytical results of section 2. Young and old individuals are net receivers of the government budget, whereas working-age individuals are net contributors, as previously shown in Figure 3. As the population ages, the share of net contributors declines, as evidenced by a rising dependency ratio, leading to an increasing deficit over time. We remark that even in this rich projection exercise, which considers several dimensions (age, education, country of birth, and gender), the dynamics of the deficit are always tightly linked to the dependency ratio.

¹⁹This net migration flow includes return migration, that is, individuals returning to their origin countries.

The Figure also compares scenarios for net migration flow, ranging from 0 to 2 percent of the 2019 total population. These are depicted along the horizontal axis of the figures. The results show a positive relation between net migration and fiscal sustainability. A smaller migration inflow leads to larger primary deficits, while larger flows result in smaller deficits. This is mainly due to the age structure of net migration flows being more concentrated in working-age groups than in the resident population, as shown in Table 1. For this reason, a more intense inflow of migrants leads to lower dependency ratios and, therefore, smaller fiscal deficits.

The relationship between net migration flows and primary deficits is not linear. As panel (a) also shows, the effect of intensifying immigration flows on the deficits is positive, but diminishing, while constraining migration has increasing effects: the fiscal costs of “building walls” increase with the size of the restriction. By 2100, a smaller immigration flow by 0.4 p.p. (i.e., a shutdown of the net migration flow) would lead to a deficit larger by about 4.1 p.p., while a larger immigration flow by 0.4 p.p. (i.e., doubling the net migration flow) would only reduce the 2100 deficit by 1.5 p.p. This convexity is rooted in population dynamics, as shown in Proposition 1.

4.2.2 Implications for the Fiscal Burden of Aging: a Convex Policy Frontier

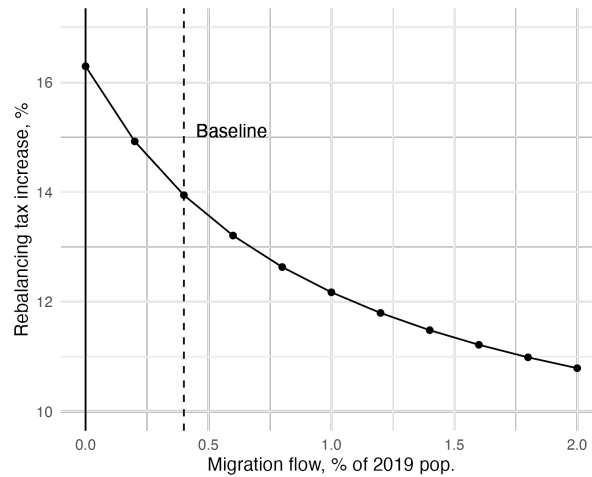
The interpretation of these dynamics is clear. As fertility is below replacement, the population of Europe is undergoing a transition towards a “stationary through immigration” population, i.e., one that is sustained only by a regular inflow of migrants. Increasing the flow of immigrants accelerates that transition, but there is a limit to how much it can accelerate. This is why immigration alone cannot solve the fiscal burden of an aging population and, conversely, why the “costs of building walls” are rising. As the share of the immigrant population increases over time, this subpopulation’s age structure plays an increasingly important role in driving the overall primary deficit. Since the immigrant population has a lower dependency ratio, this helps shrink the deficit. As a result, stronger net migration inflows lead to smaller deficits, but each additional increment in inflows has a diminishing impact on deficits.

This mechanism explains why many previous studies have concluded that immigration cannot fully resolve the public finance sustainability issues faced by developed economies (see Preston, 2014, for a review). Such studies, following the standard procedure in official demographic projections, also assumed a constant immigration inflow.

We now examine the rebalancing tax increase, θ_τ , across scenarios with different net migration levels. Figure 5 draws the frontier between the net migration flow and θ_τ ,

here shown as the EA average weighted by country potential GDP. The baseline scenario obtains a θ_τ of 14 percent. This means an additional 5.9% of GDP in revenues each year, compared with the current 42% of GDP collected in taxes and social contributions. This tax adjustment becomes more severe with a restriction to immigration flows, as shown in Figure 4.

Figure 5: Frontier between the Level of Migration and the Rebalancing Tax Increase, θ_τ .



Note: The figure shows the rebalancing tax increase, θ_τ , for the different net migration scenarios. This metric is the weighted average of the country-specific rebalancing tax increase metrics across the EA countries, with 2019 potential GDP as weights. The dashed line indicates the size of migration flows in the baseline scenario.

Source: Authors' calculations.

The chart also shows that the relationship between net migration and the fiscal burden of aging is convex. The gains from expanding migration flows are highest at low levels, and then decline—in this sense, increasing migration has diminishing returns for public finances. A scenario where migration flows are totally shut down increases θ_τ by 2.3 percentage points, from 14% to 16.3%, or 1 percent of GDP. Conversely, if future immigration flows doubled, from the current 0.4% to 0.8% of the 2019 population, the rebalancing tax adjustment would fall by only 1.4 percentage points. It follows that immigration cannot, per se, eliminate the fiscal burden of aging. As the Figure shows, after increasing net migration flows to 2%, a level well above any historical migration experience, fiscal sustainability would still require a 10.8 percent rebalancing tax increase, and further increases would obtain very small gains.²⁰ Table 4 shows the exact figures for the rebalancing tax increase.

The key takeaway is that maintaining immigration flows can be an important aid to Europe in addressing the fiscal burden of an aging population. Shutting down net migration flows would require an additional permanent tax increase of 1% of the EA's

²⁰A recent report also points in this direction, indicating that the scale required to offset population aging substantially exceeds historical experience (European Bank for Reconstruction and Development, 2025).

Table 4: Imbalance Metrics in Different Migration Scenarios.

Size of net migration flow	θ_τ	Diff.	in % pot. GDP
0.0% of 2019 pop. (shut-down)	16.3%	–	–
0.4% of 2019 pop. (baseline)	14.0%	-2.3 p.p.	-1.0
0.8% of 2019 pop. (doubling)	12.6%	-1.4 p.p.	-0.6

Note: The value of θ_τ reported corresponds to the weighted average of the rebalancing tax increase of each country computed according to Equation (6), weighted by the potential GDP of 2019 of each EA country. The table reports the metric for three net migration scenarios: shutdown of migration flows (0%), baseline migration flows (0.4%), and doubling of migration flows (0.8%).

Source: Authors' calculations.

potential GDP each year, on top of the 5.9% baseline, to restore fiscal sustainability. It corresponds to an annualized value of €6,048 for the average native 30-year-old worker, €845 higher than the €5,203 with baseline immigration. Table E.1 in the Appendix shows, from an individual perspective, this tax increase, for the EA average and at the country level.

General Equilibrium Effects. A caveat in our results concerns possible general equilibrium effects of immigration. Immigration expands the labor force, which tends to reduce wages and raise interest rates due to an increased labor-to-capital ratio, reducing labor tax revenues and increasing the cost of servicing public debt. However, Colas and Sachs (2024) argue that such effects may be mitigated in the presence of complementarity between low- and high-skill workers. Since most immigrants are low-skilled, their presence can boost the productivity of high-skilled natives and, due to tax progressivity, disproportionately increase their wages and tax contributions, partially offsetting the negative general-equilibrium effects. Indeed, the authors show that these indirect fiscal gains outweigh the direct fiscal costs in their US exercise.

Similarly, Busch et al. (2020) find that the general equilibrium effects of immigration can be modest in a similar setting. Analyzing the large influx of refugees to Germany, they report small net welfare gains that grow over time: a 0.5% welfare increase in 2020 and a 1% welfare increase by 2060. These gains persist despite downward pressure on wages, suggesting that the main channels operate through demographic change: the influx of migrants increases the working-age population, which in turn lowers the dependency ratio and the fiscal cost-to-benefit ratio, thereby alleviating the tax burden on native workers.

Another potential source of general equilibrium effects is the positive response of immigration flows to improved fiscal conditions in the destination country, which could bias the effects of immigration downward. While this is theoretically possible, our analysis treats migration flows as exogenous. Moreover, long-term fiscal conditions

in host countries are unlikely to be a primary determinant of migration decisions, especially when compared to more immediate factors such as current levels of taxes or socio-economic conditions in the origin country, linguistic ties, and geographic proximity (Mayda, 2010; Grogger and Hanson, 2011).

4.3 Is Fertility an Alternative to Migration?

Policies promoting fertility are often seen as an alternative to migration for addressing the fiscal burden of aging. We consider this option within our framework in an exercise that varies native fertility levels, as shown in Table 5. We take the zero net migration scenario as our starting point, which features the observed level of native fertility, around 1.6 children per woman (Scenario A in the table). We then compare this with two alternatives: first, a higher fertility rate scenario, where we set it at the replacement level, 2.1 children per woman, and keep net migration at zero (Scenario B);²¹ Second, we keep fertility at around 1.6 children per woman, and set net migration flows at their 2019 levels (Scenario C, also our baseline in the main results above).

In the first alternative considered (B), when native fertility is higher, the rebalancing tax increase is almost unchanged, only decreasing by 0.7 percentage points. In contrast, when we increase the net migration flow to 2019 levels (Scenario C), the rebalancing tax increase falls by 2.3 percentage points compared to Scenario A (as previously shown in Figure 5). The high-fertility scenario generates higher dependency ratios and, therefore, projected primary deficits than the no-migration scenario during the first 40 years of the projection. This effect is compensated for by a lower long-run primary deficit.

Table 5: Imbalance Metrics in High Native Fertility Scenario

Scenario	Native fertility	Net migration flow	θ_τ
A	Baseline (1.6 in 2020+)	0.0% of 2019 pop.	16.3%
B	High (2.1 in 2020+)	0.0% of 2019 pop.	15.6%
C	Baseline (1.6 in 2020+)	0.4% of 2019 pop. (baseline)	14.0%

Note: The value of θ_τ reported corresponds to the weighted average of the rebalancing tax increase of each country computed according to Equation (6), weighted by the potential GDP of 2019 of each EA country. The table reports the metric for scenarios with high fertility (B) and positive net migration (C), and compares them with scenario (A), where fertility and migration are low.

Source: Authors' calculations.

Intuitively, in the short run, higher fertility only increases the population share of children, who have a negative net contribution to the budget. This increase would coincide with the share of the old-age population growing as the large “baby boomer” cohort enters retirement. The benefits of higher fertility, in terms of a larger share of

²¹Note that this is a highly optimistic scenario regarding the potential of policies to increase fertility. Even if policies were successful, fertility is a slow-moving variable.

the working-age population, only come far later. For these reasons, fertility is not an alternative to migration as an instrument to moderate the increase in the dependency ratio and, therefore, the fiscal burden of aging.

4.4 Robustness Exercises

We conduct four robustness exercises. The details of these four exercises can be found in Appendix D.

Interest Rate and Productivity Growth Rate. In our baseline exercise, we set the productivity growth rate, γ , the nominal interest rate, i , and the inflation rate, π , equal to the historical average of these variables between 1995 and 2021, i.e., 0.7%, 3.8%, and 1.68%, respectively. This implies a discount factor $D = 0.986$. Given that our main results rely on projected values, we tested whether the convexity between migration flow size and the rebalancing tax increase depends on the assumption for D . In the Appendix, we compute the rebalancing tax increase for different values of D , and plot the frontier between θ_τ and the level of migration in each case. We show that the convexity holds for the different cases considered.

Fertility Rate of Immigrants. All the results shown so far rely on the same assumption about the fertility rate of first-generation immigrants, taken from the data as 2.35 (on average across countries and ages) in 2019 and assumed to decline slowly, converging towards 2.1 children per woman by 2100. This assumption could affect the measured fiscal benefits of migration. In the Appendix, we test for that by computing the rebalancing tax increase with different assumptions for the immigrant fertility rate,²² as well as the frontier between θ_τ and the level of migration. Convexity is observed across these scenarios.

Immigrants' Offspring Fertility. In the baseline exercise, the second and later generations of immigrants exhibit the same demographic behaviors as the native population. An alternative modeling choice is to set immigrant offspring fertility equal to that of the first-generation immigrants. In the Appendix, we present the results of an alternative exercise in which second-generation immigrants have the same fertility as their immigrant parents. Fiscal balances are slightly worse with higher fertility at any level of

²²Across these scenarios we keep the assumption that 2nd-generation immigrants have the same fertility as natives.

immigration flows, but the differences are subdued. We also observe that the convex relationship between the rebalancing tax increase and the net migration inflow level remains.

Education Composition of Immigrants. Our baseline exercise assumes that the educational composition of new immigration is the same as that observed in 2019. Differences in the composition of education change the total net benefits of immigration. In the Appendix, we show, however, that the rebalancing tax increase still exhibits decreasing returns to net immigration flows when all immigrants have completed tertiary education or have only completed primary education.

5 Immigration and the Fiscal Burden by Country

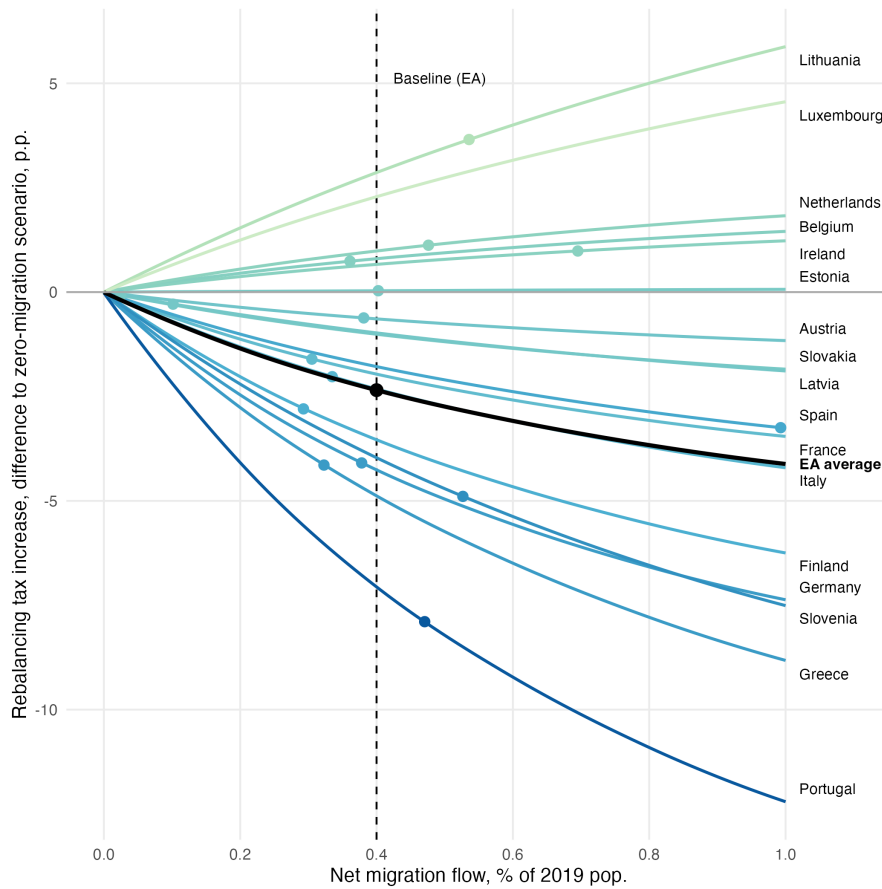
In the European Union, migration policy is mostly decided at the national level. In recent years, the European Commission has put forward some measures toward a common migration policy. In this section, we analyze how the fiscal burden of aging differs across the EA countries and how changes in net migration flows affect it. This exercise reveals policy-relevant cross-country differences, enabling us to assess the feasibility of a common migration policy. Beyond that, the exercise sheds light on how our model framework and main conclusions apply to specific country cases.

Figure 6 shows the policy frontiers between migration flows and the rebalancing tax increase faced by each country. We plot how much θ_τ changes in comparison to a closed-borders scenario for each country (gradient of colors) and for the EA average (black line).²³ The dots on each frontier mark the net migration level of each country observed in 2019, which is kept in the baseline scenario.

Immigration relieves the fiscal burden of aging in most countries. Higher net migration flows reduce the rebalancing tax increase across France, Germany, Italy, and Spain, thereby driving the EA aggregate results. This is not based on extreme immigration flows. With the notable exception of Spain, most countries range between 0.3 and 0.5% of the total population, around the cross-country average. In general, in these countries, immigrants from outside the EU arrive between the ages of 20 and 30, and out-migration is relatively muted. The conditions in our model apply, and, as with the EA average, they reduce the tax increase required of all taxpayers, even though they make smaller per

²³Note this is slightly different from the preceding Figure 5, with the EA-level frontier, which showed θ_τ directly for each level of migration. The change is intended to facilitate comparison, given that baseline fiscal conditions differ significantly across countries.

Figure 6: Rebalancing Tax Increase, Difference to Zero-Migration Scenario for each EA Country



Note: The figure shows, for different scenarios of net migration, the difference in the rebalancing tax increase, θ_T , compared to the zero-migration case, in the different countries of the EA. The dots on each frontier point to the baseline scenario in each country, which is based on the net migration flows observed in 2019. The baseline case for the EA aggregate is further marked by a vertical dashed line. The baseline flow for Luxembourg is not plotted as it was close to 2% of the 2019 population.

Source: Authors' calculations.

capita contributions. The cross-country results make it clear that the key comparison is between immigrants' lifetime net contribution from the age of entry and natives' lifetime net contribution at birth. Figure E.2 in the Appendix shows the former is larger in most countries, where immigrants alleviate—though not eliminate—the fiscal burden of aging for all.

However, for a small set of countries, larger net migration flows require a greater increase in the rebalancing tax. This can happen for two reasons. In some countries, immigrants make substantially lower contributions than natives, including in the working ages. In these cases, even if they enter at ages 20-30, they make a larger negative net contribution over their lifetimes than natives starting from birth. This is the case in Belgium, the Netherlands, and Ireland. In other words, in these countries, working-age immigrants

entering the population have a similar, if not worse, effect on fiscal sustainability than newborns, given current tax-benefit policies; as such, the demographic mechanism does not apply. In a couple of other countries, there is also a negative relationship between immigration and fiscal balances, but for a more straightforward reason. In Lithuania and Estonia, the arrival age of most immigrants is later than 30, so they miss some years of positive net contributions, which brings their lifetime contribution below that of a newborn native. Our model is based on the common case that immigrants arrive at the *early* working ages. The case of these two countries illustrates how the sign of the fiscal effects of immigration may change when this does not apply.

Additionally, the Figure shows that for countries where net migration is beneficial, increasing inflows has different effects on the rebalancing tax increase. These differences are a consequence of the diminishing returns of migration that we uncover in this paper. Countries face different costs and benefits of immigration depending on where they sit on the convex frontier. Spain, which has high migration flows, has little to gain from further increases and could even allow them to retreat with little impact on the fiscal burden of aging. Finland, Greece, or Germany, instead, would face significantly higher per-capita fiscal costs if they were to cut back on immigration. Portugal would lose even more, as it is an outlier—the only country in our data in which immigrants are estimated to make larger net contributions than natives. Countries like Austria or Slovakia are on the opposite end, where immigrants make significantly lower net contributions than natives, and only provide a small relief to the fiscal burden of aging. These results highlight how challenging it is to set a common migration policy within the European Union. Restricting immigration flows, as currently under discussion in many countries, will have very heterogeneous impacts on public finances across countries. Our results are also relevant when thinking about refugee allocation programs, since some countries could benefit, from a fiscal perspective, by receiving more migrants.

Although not the central focus, our framework also sheds light on understanding which countries' public finances are more vulnerable to aging. Figure E.1 in the Appendix decomposes the differences in the baseline rebalancing tax increase across countries between demographic and public finance macro sources. This decomposition shows, for instance, that although the baseline rebalancing tax increases in Germany and France are very similar, France is mostly due to current aggregate imbalances (high debt and a large primary deficit), whereas Germany is mostly due to demographic reasons, with a budget structure that is vulnerable to demographic change in the country.

6 Conclusion

In this paper, we revisited the question of the fiscal effects of immigration, focusing on a specific channel: how immigration flows change the demographic composition of low-fertility societies. We showed that immigration generates a negative, convex relationship between the intensity of immigration flows and the fiscal burden of aging. Restricting immigration raises future dependency ratios and fiscal imbalances at an increasing rate, while expanding immigration reduces them, but with diminishing returns.

This mechanism does not rely on immigrants' skills or on favorable labor-market effects. Even when immigrants have lower lifecycle net fiscal contributions than natives, immigration can improve the fiscal position because migrants arrive disproportionately at working ages, after childhood and before old age. In that case, their remaining lifetime contributions can exceed those of native newborns, so immigration relieves the fiscal burden on resident taxpayers despite migrants' lower average skill levels.

Quantitatively, we found that maintaining 2019 immigration flows lowers the fiscal burden of aging in the EA from 16.3 percent to 14.0 percent. This means that shutting down immigration would require an additional fiscal adjustment of about 1 percent of GDP per year to restore long-run fiscal balance. By contrast, raising fertility among natives is much less effective at alleviating the fiscal burden of aging, because newborns increase dependency in the short run and contribute only much later in life. The magnitude of these effects varies substantially across EA countries, reflecting differences in immigration flows, demographic structures, and tax-benefit systems, implying that a uniform EU migration policy may not be optimal.

More broadly, our results imply that in low-fertility societies, where dependency ratios would otherwise rise rapidly, immigrants can relieve the fiscal burden of aging for natives by rejuvenating the population, even when they are predominantly low-skilled. In such settings, the fiscal costs of restricting immigration can be large.

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A Population Projection Model

The model described in section 2 is a linear differences model, which we can write as:

$$\mathbf{P}_t^i = A\mathbf{P}_{t-1}^i + m^i, i = \{N, F\} \quad (8)$$

where $\mathbf{P}_t^i = \begin{bmatrix} P_t^{Y,i} & P_t^{W,i} & P_t^{O,i} \end{bmatrix}'$, $m^i = \begin{bmatrix} 0 & M & 0 \end{bmatrix}'$, with $M = 0$ for $i = N$. The transition matrix A is given by

$$A = \begin{bmatrix} 1 - \pi_w & f & 0 \\ \pi_w & 1 - \pi_o & 0 \\ 0 & \pi_o & 1 - \pi_m \end{bmatrix},$$

where π_m is the probability of death, π_o is the probability of entering retirement, π_w is the probability of entering working age, and f is the fertility rate. Further label the upper-left block of A as:

$$B \equiv \begin{bmatrix} 1 - \pi_w & f \\ \pi_w & 1 - \pi_o \end{bmatrix}.$$

Define total population vectors $P_t = (P_t^Y, P_t^W, P_t^O)'$, where for each age group a , $P_t^a \equiv P_t^{a,N} + P_t^{a,F}$. Let $u = (1, 0, 1)'$ be an auxiliary vector that selects old and young populations, and $v = (0, 1, 0)'$ be a vector that selects the working-age population. The dependency ratio is

$$D_t = \frac{u'P_t}{v'P_t} = \frac{P_t^O + P_t^Y}{P_t^W}.$$

Write immigrant inflows as $m_t = (0, M_t, 0)' = M_t v$.

Throughout, we maintain the following assumptions:

- **A1:** $f < \pi_o$. This defines sub-replacement fertility in this model.
- **A2:** $\pi_w + \pi_o < 1$. Individuals remain in each stage for multiple periods.
- **A3:** $\rho(B) > 1 - \pi_m$, where B is the upper-left block of A governing the young and working-age groups. This demands that the combined persistence of the young and working-age groups exceed the survival rate of the old group.
- **A4:** $\pi_m \leq \pi_w$. This ensures old-age mortality is no greater than the rate at which the young reach working age.

A.1 Proof of Lemma 1.

Path to Extinction. Under zero migration, the system of difference equations that characterizes the model reduces to $\mathbf{P}_t = A\mathbf{P}_{t-1}$. The solution is given by $P_t = A^t P_0$. To show that the population shrinks to extinction, it suffices to show that the eigenvalues of A are less than 1. Note that A is block lower-triangular:

$$\begin{bmatrix} B & \mathbf{0} \\ a^T & 1 - \pi_m \end{bmatrix},$$

where $a^T \equiv (0, \pi_o)$. Since the eigenvalues of a block lower-triangular matrix are the union of the eigenvalues of its diagonal blocks, the spectral radius of A satisfies

$$\rho(A) = \max\{1 - \pi_m, \rho(B)\}.$$

By definition $\pi_m \in (0, 1)$, so we need only to show that $\rho(B) < 1$. To that end, evaluate the characteristic polynomial, $p_B(\lambda) \equiv \det(\lambda I - B)$, which is U-shaped, at $\lambda = 1$:

$$p_B(1) = 1 - \text{tr}(B) + \det(B) = \pi_w(\pi_o - f).$$

Under A1, $p_B(1) > 0$. Moreover, it follows from A1 and A2 that $p_B(0) = \det(B) = (1 - \pi_w)(1 - \pi_o) - f\pi_w > 0$, and $\text{tr}(B) = 2 - \pi_w - \pi_o \in (0, 2)$. Thus, both roots of p_B are real and belong to $(0, 1)$, which implies that $\rho(B) < 1$ and the population is on the path to extinction.

Dependency Ratio Convergence. To show convergence, we exploit the block structure of A . The 2×2 sub-block B is itself primitive: both off-diagonal entries $f, \pi_w > 0$ ensure irreducibility, and the positive diagonal entries ensure aperiodicity. By the Perron-Frobenius theorem, there exists a unique (up to scaling) strictly positive right eigenvector $(r_Y, r_W)' \gg \mathbf{0}$ associated with $\rho(B)$.

Under A3, the dominant eigenvalue of A is $\lambda_1 \equiv \rho(B)$. Solve $A\mathbf{r} = \lambda_1\mathbf{r}$ with $\mathbf{r} = (r_Y, r_W, r_O)'$. The upper block gives $B(r_Y, r_W)' = \lambda_1(r_Y, r_W)'$, which, as just noted, yields $(r_Y, r_W) \gg \mathbf{0}$. The third row gives $\pi_o r_W + (1 - \pi_m)r_O = \lambda_1 r_O$, so that

$$r_O = \frac{\pi_o r_W}{\lambda_1 - (1 - \pi_m)} > 0,$$

since $\lambda_1 > 1 - \pi_m$ and $r_W, \pi_o > 0$. Hence $\mathbf{r} \gg \mathbf{0}$. Because $\lambda_1 = \rho(A)$ is a simple

eigenvalue that is strictly larger in modulus than every other eigenvalue of A , the standard spectral decomposition of A^t gives

$$\rho(A)^{-t} A^t \longrightarrow \frac{\mathbf{r} \ell'}{\ell' \mathbf{r}},$$

where ℓ is the left eigenvector associated with λ_1 . It follows that for any initial condition with $P_0^Y, P_0^W > 0$

$$\lim_{t \rightarrow \infty} \frac{P_t}{\|P_t\|} = \frac{\mathbf{r}}{\|\mathbf{r}\|}.$$

Since $\mathbf{r} \gg \mathbf{0}$, the dependency ratio converges to a finite, positive limit:

$$\lim_{t \rightarrow \infty} D_t^{\text{ext}} = \frac{u' \mathbf{r}}{v' \mathbf{r}} = \frac{r_Y + r_O}{r_W} \equiv D_{\infty}^{\text{ext}}. \quad \square$$

A.2 Proof of Lemma 2.

Fixing $M_t = M > 0$, the recursion $P_t = AP_{t-1} + Mv$ has fixed point

$$P = (I - A)^{-1} Mv = M \sum_{k=0}^{\infty} A^k v,$$

so $P_t \rightarrow P$ from any P_0 . Note that we can guarantee the existence of $(I - A)^{-1}$ since $\rho(A) < 1$, as shown in Lemma 1. The limiting dependency ratio is

$$D_{\infty}^{\text{SI}} = \frac{u' P}{v' P} = \frac{\sum_{k \geq 0} u' A^k v}{\sum_{k \geq 0} v' A^k v} = \frac{\pi_o}{\pi_m} + \frac{f}{\pi_w}$$

which is independent of M . □

A.3 Proof of Proposition 1.

To prove Proposition 1, we use an auxiliary lemma.

Lemma 3. (*Cohort aging.*) Under A1–A4, the dependency ratio of a working-age-entry cohort is strictly increasing: $D(A^{k+1}v) > D(A^k v)$ for all $k \geq 0$.

Proof of Lemma 3.

Write $A^k v = (Y_k, W_k, O_k)'$ and normalize $y_k \equiv Y_k/W_k$, $o_k \equiv O_k/W_k$, so that $D(A^k v) = y_k + o_k$. The law of motion $A^{k+1} v = A(A^k v)$ gives

$$y_{k+1} = \frac{(1 - \pi_w)y_k + f}{\pi_w y_k + 1 - \pi_o}, \quad o_{k+1} = \frac{\pi_o + (1 - \pi_m)o_k}{\pi_w y_k + 1 - \pi_o}.$$

Let $(a^*, 1, b^*)'$ be the Perron eigenvector of A normalized to unit working-age component. Solving $A(a^*, 1, b^*)' = \rho_1(a^*, 1, b^*)'$ with $\rho_1 \equiv \rho(B)$ yields

$$\rho_1 = 1 - \pi_o + \pi_w a^*, \quad a^* = \frac{f}{\rho_1 - 1 + \pi_w} > 0, \quad b^* = \frac{\pi_o}{\rho_1 - 1 + \pi_m} > 0,$$

where $b^* > 0$ uses A3. Expanding $D(A^{k+1} v) - D(A^k v)$ and clearing the positive denominator $W_k W_{k+1}$ shows that its sign equals that of

$$\phi(y_k, o_k), \quad \phi(y, o) \equiv (f + \pi_o) - (\pi_w - \pi_o)y - \pi_w y^2 + o(\pi_o - \pi_m - \pi_w y).$$

It therefore suffices to show $\phi(y_k, o_k) > 0$ for every finite k . We proceed in three steps; the geometric idea is that the trajectory (y_k, o_k) always sits strictly below its steady state (a^*, b^*) , where ϕ is positive.

Step 1: y_k rises strictly toward a^ .* The map $h(y) \equiv \frac{(1 - \pi_w)y + f}{\pi_w y + 1 - \pi_o}$ has derivative $h'(y) = \det B / (\pi_w y + 1 - \pi_o)^2 > 0$, since $\det B = (1 - \pi_w)(1 - \pi_o) - f\pi_w > 0$ under A1–A2. Its unique positive fixed point is a^* , and $h(y) - y$ has the sign of the downward parabola $f - (\pi_w - \pi_o)y - \pi_w y^2$, which is positive on $[0, a^*)$. Starting from $y_0 = 0$, induction gives $0 = y_0 < y_1 < \dots < a^*$. In particular,

$$\pi_w (a^*)^2 + (\pi_w - \pi_o)a^* - f = 0. \quad (*)$$

Step 2: an invariant region. Define the triangle

$$\mathcal{R} \equiv \{(y, o) : 0 \leq y \leq a^*, 0 \leq o, a^* o \leq b^* y\},$$

i.e. the old-to-young ratio never exceeds its steady-state value b^*/a^* . Clearly $(y_0, o_0) = (0, 0) \in \mathcal{R}$. Suppose $(y_k, o_k) \in \mathcal{R}$; Step 1 gives $y_{k+1} \in [0, a^*]$ and $o_{k+1} \geq 0$ is immediate. The binding requirement $a^* o_{k+1} \leq b^* y_{k+1}$, after substituting the laws of motion and the

identities for a^*, b^* , is equivalent to

$$(1 - \pi_m)\zeta_k - (1 - \pi_w)\eta_k \leq \pi_w - \pi_m, \quad \eta_k \equiv y_k/a^*, \zeta_k \equiv o_k/b^*,$$

where the hypothesis reads $0 \leq \zeta_k \leq \eta_k \leq 1$. Hence

$$(1 - \pi_m)\zeta_k - (1 - \pi_w)\eta_k \leq (1 - \pi_m)\eta_k - (1 - \pi_w)\eta_k = (\pi_w - \pi_m)\eta_k \leq \pi_w - \pi_m,$$

where the first inequality uses $\zeta_k \leq \eta_k$ and the last uses $\eta_k \leq 1$ together with $\pi_w - \pi_m \geq 0$, which is exactly A4. Thus $\mathcal{R}$ is forward-invariant, and $(y_k, o_k) \in \mathcal{R}$ for all k .

Step 3: $\phi > 0$ on $\mathcal{R}$, except at (a^*, b^*) . Since ϕ is affine in o for fixed y , it suffices to verify positivity on the boundary of $\mathcal{R}$.

- *Bottom edge* ($o = 0$): $\phi(y, 0) = (f + \pi_o) - (\pi_w - \pi_o)y - \pi_w y^2$ is a downward parabola with $\phi(0, 0) = f + \pi_o > 0$ and, by (*), $\phi(a^*, 0) = \pi_o > 0$; hence positive on $[0, a^*]$.
- *Slanted edge* $(y, o) = (ta^*, tb^*)$, $t \in [0, 1]$: writing $\phi(ta^*, tb^*) = c_0 + c_1 t + c_2 t^2$ with $c_0 = f + \pi_o$, $c_2 = -\pi_w a^*(a^* + b^*)$, and $c_0 + c_1 + c_2 = \phi(a^*, b^*) = 0$ (as $(a^*, 1, b^*)$ is an eigenvector), one obtains

$$\phi(ta^*, tb^*) = (1 - t)[(f + \pi_o) + \pi_w a^*(a^* + b^*)t] > 0 \quad \text{for } t \in [0, 1),$$

vanishing only at $t = 1$, i.e. at (a^*, b^*) .

- *Right edge* ($y = a^*$): using (*) and the eigenvector identity $\pi_o - \pi_m - \pi_w a^* = -\pi_o/b^*$, $\phi(a^*, o) = \pi_o(1 - o/b^*) > 0$ for $o \in [0, b^*)$.

Affineness in o extends positivity to all of $\mathcal{R} \setminus \{(a^*, b^*)\}$.

Conclusion. By Step 1, $y_k < a^*$ for every finite k , so $(y_k, o_k) \neq (a^*, b^*)$; by Step 2, $(y_k, o_k) \in \mathcal{R}$; by Step 3, $\phi(y_k, o_k) > 0$. Hence $D(A^{k+1}v) > D(A^k v)$ for all $k \geq 0$. $\square$

Proof of Proposition 1.

With constant inflow $M \geq 0$ the population at horizon t is $P_t(M) = A^t P_0 + M \phi_t$, where $\phi_t \equiv \sum_{k=0}^{t-1} A^k v$, so that

$$D_t(M) = \frac{u' P_t(M)}{v' P_t(M)} = \frac{\alpha + Ma}{\beta + Mb}, \quad \alpha \equiv u' A^t P_0, \beta \equiv v' A^t P_0, a \equiv u' \phi_t, b \equiv v' \phi_t.$$

Here $a, b > 0$ since $\phi_t \geq v$ has a strictly positive working-age component. Differentiating,

$$\frac{\partial D_t}{\partial M} = \frac{a\beta - b\alpha}{(\beta + Mb)^2}, \quad \frac{\partial^2 D_t}{\partial M^2} = \frac{2b(b\alpha - a\beta)}{(\beta + Mb)^3}.$$

Both signs are governed by the same quantity. Define

$$\Psi(P_0) \equiv b\alpha - a\beta = (b u' - a v') A^t P_0;$$

then D_t is strictly decreasing and strictly convex in M if and only if $\Psi(P_0) > 0$. Note that Ψ is a *linear* functional of P_0 (the scalars a, b do not depend on P_0). Writing any admissible $P_0 = Y_0 e_y + W_0 v + O_0 e_o$ with non-negative, not-all-zero coordinates,

$$\Psi(P_0) = Y_0 \Psi(e_y) + W_0 \Psi(v) + O_0 \Psi(e_o).$$

It therefore suffices to show $\Psi \geq 0$ on each basis vector, with strict positivity on each. Throughout, write $U_j \equiv u' A^j v$, $V_j \equiv v' A^j v > 0$, and $D_j \equiv U_j / V_j = D(A^j v)$, so that $a = \sum_{k=0}^{t-1} U_k$ and $b = \sum_{k=0}^{t-1} V_k$.

Case $P_0 = e_o$ (initial old). A pure-old cohort neither works nor reproduces: $A^t e_o = (1 - \pi_m)^t e_o$, so $u' A^t e_o = (1 - \pi_m)^t$ and $v' A^t e_o = 0$. Hence $\Psi(e_o) = b(1 - \pi_m)^t > 0$.

Case $P_0 = v$ (immigrant worker). Here $\Psi(v) = b U_t - a V_t$, which is positive iff $D(A^t v) = U_t / V_t$ exceeds $D(\phi_t) = a/b$. The latter is a worker-share-weighted average of $D_0, \dots, D_{t-1}$, hence bounded above by its largest term D_{t-1} . By Lemma 3, $D_{t-1} < D_t$, so

$$D(\phi_t) \leq D_{t-1} < D_t = D(A^t v),$$

giving $\Psi(v) > 0$.

Case $P_0 = e_y$ (native young). Iterating $A e_y = (1 - \pi_w) e_y + \pi_w v$ gives the decomposition

$$A^t e_y = (1 - \pi_w)^t e_y + \pi_w \sum_{j=0}^{t-1} (1 - \pi_w)^{t-1-j} A^j v,$$

i.e. a still-young remainder plus a geometrically weighted history of past worker graduations. Applying u' and v' (with $u' e_y = 1$, $v' e_y = 0$) and writing $g_j \equiv (1 - \pi_w)^{t-1-j}$,

$$\alpha = (1 - \pi_w)^t + \pi_w \sum_j g_j U_j, \quad \beta = \pi_w \sum_j g_j V_j.$$

Therefore

$$\Psi(e_y) = b\alpha - a\beta = (1 - \pi_w)^t \sum_k V_k + \pi_w \sum_{j,k} g_j (U_j V_k - V_j U_k).$$

Symmetrizing the double sum over the pair (j, k) and using $U_j V_k - V_j U_k = V_j V_k (D_j - D_k)$,

$$\sum_{j,k} g_j (U_j V_k - V_j U_k) = \sum_{j < k} (g_j - g_k) V_j V_k (D_j - D_k).$$

For $j < k$: $g_j - g_k < 0$ because $1 - \pi_w \in (0, 1)$ carries a higher exponent at j ; and $D_j - D_k < 0$ by Lemma 3. The product of the two negatives is positive, so every term is positive, and the double sum is ≥ 0 (strictly so for $t \geq 2$). Combined with the strictly positive leading term,

$$\Psi(e_y) > 0.$$

Conclusion. Each of $\Psi(e_y), \Psi(v), \Psi(e_o)$ is strictly positive, so for any admissible $P_0 \neq 0$, $\Psi(P_0) > 0$. Hence $\partial D_t / \partial M < 0$ and $\partial^2 D_t / \partial M^2 > 0$ for every finite $t \geq 1$ and all $M \geq 0$; that is, $D_t(M)$ is strictly decreasing and strictly convex in M . $\square$

A.4 Proof of Corollary 1.

For $t < T^*$, inflows are constant at M , so we are in the setting of Proposition 1 and $D_t(M)$ is strictly decreasing and strictly convex in M .

For $t \geq T^*$, we have $M_t = 0$, so the population follows $P_t = AP_{t-1}$. By Lemma 1, it reverts to the extinction path: $P_t \rightarrow 0$ and $D_t \rightarrow D_\infty^{\text{ext}}$. $\square$

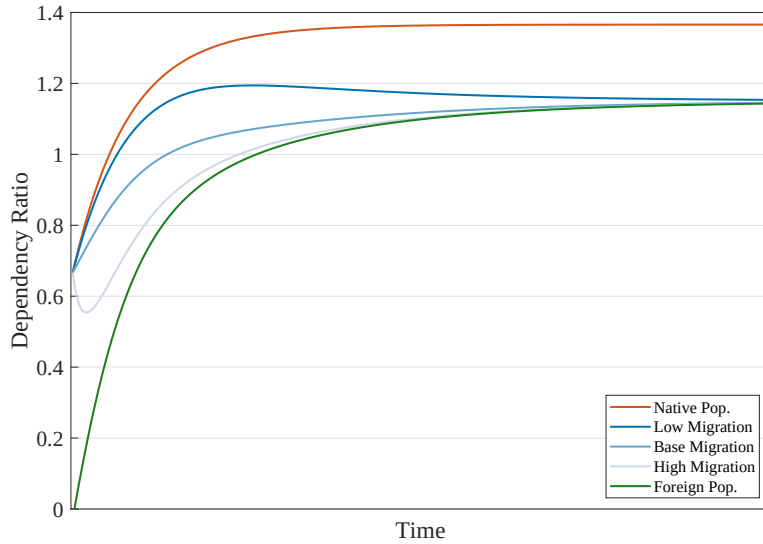
A.5 Simulation Results.

We illustrate these dynamics by simulating the model. We choose the following values for the parameters: $\pi_m = 0.035$, $\pi_o = 0.025$, $\pi_w = 0.04$, and $f = 0.012$. For the first generation of migrants, we set fertility to 0.03; migrants' descendants have fertility equal to that of natives. These numbers are not intended to represent any specific country but rather reflect the facts observed in most European countries: an aging native population with a steady inflow of working-age immigrants, with the first generation of migrants having a high fertility rate. Importantly, these parameters lie within the parameter space implied by assumptions A1–A4.

We examine the transition in the dependency ratio across three immigration intensities:

low, medium, and high. Figure A.1 plots the dependency ratio along the transition for the different scenarios.

Figure A.1: Dependency Ratio Along the Demographic Transition.



Note: The dependency ratio is the sum of the young and old populations divided by the working-age population. The figure plots it for the native population (red), the foreign population (green), and the total population under low, medium, and high immigration intensities (shades of blue).

Source: Authors' calculations.

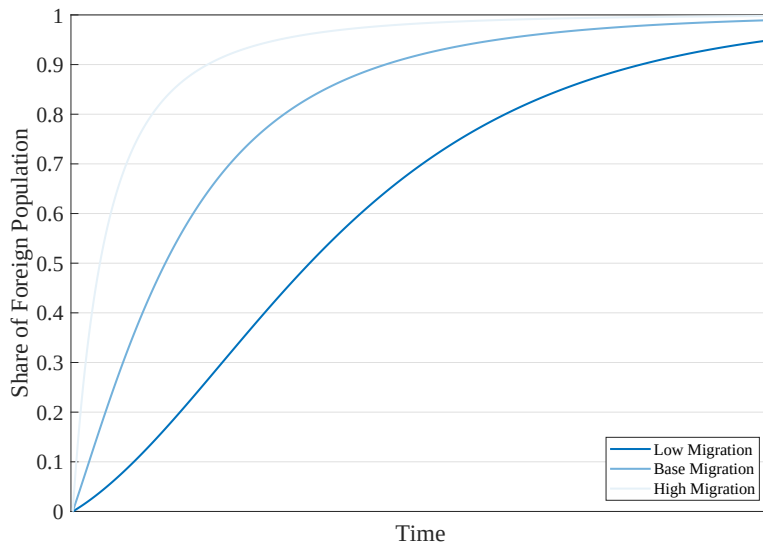
The figure illustrates the stable population results described above. Regardless of the intensity of immigration flows, the long-run population dependency ratio converges to that of the foreign group. Without immigration, the long-run dependency ratio would be equal to that of the native group, which is substantially higher.

The figure also shows that higher immigration flows lead to faster convergence to the steady-state dependency ratio. This is a direct implication of Proposition 1. Intuitively, as the share of immigrants increases over time, the dependency ratio becomes closer to the immigrant group's and farther from the native group. Figure A.2 illustrates these dynamics.

At low levels of immigration, the population dependency ratio initially follows similar dynamics to those of the native group, as the share of immigrants remains low for several periods. With higher levels of immigration, the share of immigrants rises quickly, and the population's dependency ratio quickly jumps to the immigrant group's transition path. The figure A.2 also shows that, for any given period, the share of immigrants increases non-linearly with the intensity of immigration flows, as it is converging to 1.

In summary, our simple demographic model has two implications for how immigration

Figure A.2: Share of Foreign Population Along the Demographic Transition.



Note: The figure shows the share of the foreign population along the transition. The foreign population includes immigrants and their descendants.

Source: Authors' calculations.

flows affect the population age distribution. First, the size of the immigration flow does not affect the stationary population age distribution, except in the knife-edge case of zero migration. Second, increasing immigration flows improves the dependency ratio along the transition, but at a decreasing rate.

B Methodological Details

B.1 Cyclically-Neutral Fiscal Aggregates

Our methodology involves choosing a base year, $\bar{t}$, from which demographic and macroeconomic assumptions are used to project demographic profiles over time and to compute the counterfactual government budget. Therefore, the results may depend on this choice, since projecting from $\bar{t}$ onward implicitly carries the business cycle position of the base year. To clean the long-term projections from the initial business-cycle position, the literature has proposed three solutions.

A first approach is to choose a base year in which GDP is close to potential GDP. This approach is used in Franco et al. (2020), where 2017 is chosen as the base year, which, according to the AMECO database, is the year in the last decade when the output gap in Portugal (the country the authors study) was closest to 0. A second

approach used in Feist, Raffelhuschenm, and Vanne (1999) involves departing from the most contemporaneous period and making ad hoc adjustments along the projection to a cyclically neutral state. In this paper, we use a third approach that computes cyclical-neutral fiscal aggregates for the base year.

The idea is to compute the budget item level that would be observed if the economy in period $\bar{t}$ were at full employment, and carry those values forward to the subsequent steps of the methodology. This way, we obtain cyclical-neutral fiscal aggregates. This is similar to what is done in Bonin, Patxot, and Souto (2014), building on the work of Girouard and André (2006). In order to compute the cyclical-neutral item i , $(T_{\bar{t}}^i)^*$, consider the following relationship:

$$\frac{(T_{\bar{t}}^i)^*}{T_{\bar{t}}^i} = \left(\frac{Y_{\bar{t}}^*}{Y_{\bar{t}}} \right)^{\varepsilon_{i,Y}},$$

where $T_{\bar{t}}^i$ is the observed value of the revenue item i in the base year, $Y_{\bar{t}}^*$ is the potential GDP of the year $\bar{t}$, $Y_{\bar{t}}$ is the observed value of GDP in year $\bar{t}$ and $\varepsilon_{i,Y}$ is the elasticity of revenue category i with respect to the output gap. We compute the cyclical-adjusted values for 4 of the 5 revenue budget items: PIT, CIT, VAT, and SS contributions. For the expenditure items, we consider only the unemployment benefits item, which exhibits the same relationship, mutatis mutandis. The other budget items are less subject to business cycle fluctuations. In Table C.2, we describe the sources of the elasticity numbers that we use, as well as the source of the potential GDP.

B.2 Clemens (2022) Adjustment for Capital Income Taxes

Standard partial-equilibrium fiscal accounting omits an important channel: when a firm hires an immigrant, profit maximization requires it to employ additional capital. This additional capital generates capital income—and hence additional tax revenue—that is omitted when only labor income taxes are considered. Following Clemens (2022), we adjust our estimates to account for a conservative measure of capital income taxes induced by immigrant employment.

For each country, we adjust the baseline revenue per capita attributed to new immigrants, i.e., additional immigrants entering the population after the base year 2019, by a component given by the following expression for demographic group x :

$$\Delta \tau_x^k = \left(\tau_x^{PIT} + \tau_x^{SS} \right) \times \left(\frac{\mathcal{T}^{\leq}}{\mathcal{T}_{k(x)}^L} \right) \times \frac{\alpha}{1 - \alpha} \times \omega,$$

where $\mathcal{T}^{\leq}$ denotes the tax wedge on capital income and $\mathcal{T}^{\mathcal{L}}$ on labor income (for education level k) respectively, α is the aggregate capital share of income, and ω is the share of capital owned by natives. The adjustment is performed at the country level. Data for $\mathcal{T}^{\mathcal{L}}_k$ are obtained from the OECD Taxing Wages 2024 report²⁴, while data for $\mathcal{T}^{\leq}$ come from the European Commission’s Taxation Trends in Europe 2020 report. α , the capital share of income, is calculated as one minus the labor share of income, computed from Eurostat national accounts for the same year, as the ratio of total compensation of employees (wages and salaries plus employers’ social contributions) to gross value added in each country.

We also introduce a scaling factor, ω , due to the differences in the firm and property ownership between natives and immigrants. The fraction of corporate shares owned by immigrants is smaller than that of natives; it would be unreasonable for both groups to have the same corporate income tax demographic profile. We introduce an adjustment that accounts for these differences in firm ownership rates and, in practice, allocates the burden of CIT mostly to the native group.

B.3 Building Population Projections

For the most part, we use inputs from EUROPOP2019 (such as age-specific mortality and fertility rates) and combine them with additional data and assumptions to build customized demographic projections for each country’s population by age, gender, education, and birthplace for 2019-2100. After 2100, we assume a fully stationary population.

Fertility and Mortality. The mortality rate is the same across education levels and countries of birth, differing only by age and gender.²⁵ We abstract from differences that could imply different life expectancy and mortality rates between EU-born and non-EU-born, or educational backgrounds.

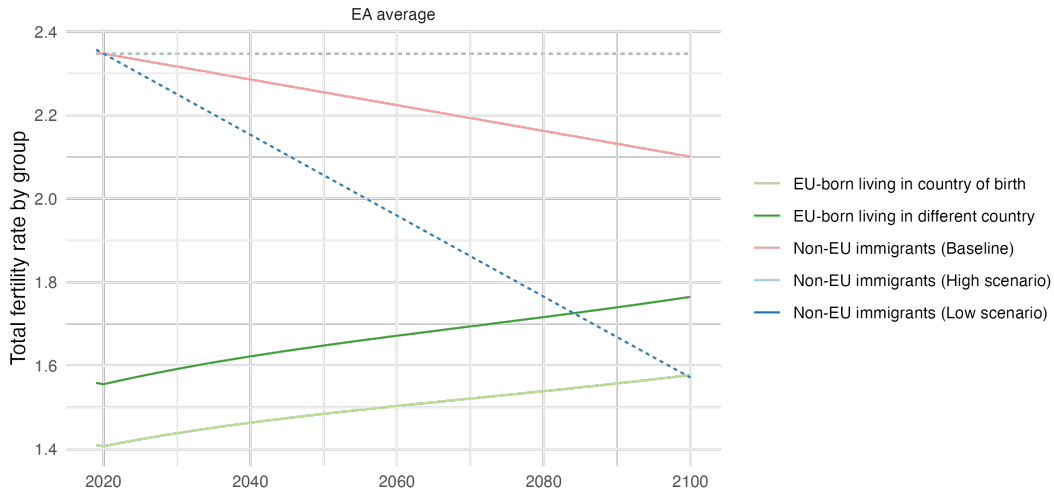
Regarding fertility rates, we use data on live births by mother’s age and country of birth for the base year, 2019. Then, for the EU-born group, we use the assumed fertility-rate growth from the EUROPOP2019 projections. This leads to a linearly increasing convergence path for all EA countries until 2100. For the non-EU-born group, we

²⁴The database provides labor tax wedges for three income groups: 2/3 of the average income, average income level, and 2/3 above average. We attribute these, respectively, to primary-, secondary-, and tertiary-level education workers

²⁵It corresponds to the mortality projected for each country in the central projection of EUROPOP2019 by age and gender, with a small modification at the age of 100, where we set the survival probability to 0.

assume a decreasing linear convergence path toward 2.1 by 2100. These fertility rates apply both to immigrants and their offspring. The time path of these total fertility rate assumptions is shown in Figure B.1, which additionally shows the fertility rates used for the robustness check scenarios described in Appendix D.3.

Figure B.1: Assumptions for the Time Path of Fertility Rates by Country of Birth Group.



Note: The plot shows the fertility rate path by country of birth group used in the demographic projections. The source of the EU-born population path is the Eurostat EUROPOP demographic projections. The Non-EU-born path is an assumption that we do and that we test for the sensitivity of it with a high fertility scenario (where we assume it to be constant and equal to the value observed in 2019) and a low fertility scenario (where we assume a linear convergent trajectory towards the projected value for natives in 2100).

Net Migration and Country of Birth. Net migration is set to remain constant at the 2019 values, and the age distribution is held at the baseline. For each age and gender, these are split by country-of-birth group, using the 2019 observed shares. We do the same for emigrants, then compute net migration as the difference between the two.

Education. We also split the population by education level. Data on education and country-of-birth composition by age group are available for 2019. We build a future path for the distribution of education using education shares observed by country of birth for the cohort aged 25 in 2019. This further relies on some assumptions for the projection: (i) for each individual, education does not fall or increase along the life cycle beyond age 25; (ii) all education paths are complete by that age; (iii) immigrants' offspring have the same education distribution as the natives.

These assumptions imply that, during the projection period, the education distribution converges to a stationary distribution. In that stationary state, the share of the working-age population with low education is smaller than in the base period, and, conversely,

the shares of the working-age population with medium and high education are higher. Our education distribution projection can be seen as conservative, since most peripheral EU countries show an increasing share of the population pursuing further education. Because it is beyond the scope of this paper to analyze the impact of different education distributions on public finances, we stick with our conservative projection of the evolution of education levels. Furthermore, significant changes in the distribution of education have implications for productivity growth, with second-order effects on taxes and benefits that may require a general equilibrium analysis.

B.4 Estimation of the Demographic Profiles

Taxes and Social Benefits. The demographic profiles for all budget items except education and healthcare are estimated using Equation (7) with the EU-SILC and the EU-HBS, provided by Eurostat, as well as the HFCS provided by the ECB. However, not all budget items and variables are available for all countries. Moreover, some countries have small sample sizes that do not allow us to estimate across all demographic groups. Here we list the deviations we do for the different countries and budget items:

- The demographic profiles of Estonia, Latvia, and Lithuania are jointly estimated. Arguably, these countries share many common features, making the benefits of increasing statistical power from joining outweigh the costs of losing this heterogeneity.
- For all countries, the profiles of the primary and the secondary education levels are jointly estimated, since for some countries the population in the primary education group is too small for a meaningful estimation.
- Spain does not report individuals' countries of birth. Hence, Spanish demographic profiles use the European average differences between countries of birth.
- The VAT demographic profiles of Austria and the Netherlands do not differ by education.
- PIT and property tax are reported at the household level. We assigned each household member a value, paid in proportion to their share of the household's total income.

Education and Healthcare. We derive an age profile of education spending from Eurostat data, which includes government spending by level of study. We allocate spending to the expected age at which an individual attends each level of study, based

on the report by Motiejunaite-Schulmeister, Sicurella, and Birch (2022). We do not consider heterogeneity by gender, education, or country of birth.

Healthcare age-gender profiles are obtained directly from the data made available by the European Commission (2021). We use the EU average for all countries.

Proportional Adjustment to Match National Accounts. The last step is to adjust the estimated profiles using a proportionality rule to match national accounts aggregates for the budget components of interest. This ensures our projection exercise is consistent with the aggregate budget balance. Note that this is necessary not only in cases where we use the distribution of proxy variables to map taxes and benefits to demographic groups (e.g., household private equity holdings to map corporate income tax). This step is also necessary for variables that can be directly mapped from survey data, because survey aggregate estimates are typically not consistent with national accounts aggregates, either due to timing or limitations of the survey data.

C Additional Data Sources Details

Table C.1: Microdata Sources.

Item	Source	Variables
PIT	EU-SILC	Income taxes (HY140G)
Property tax	HFCS	Real estate holdings (da1400)
VAT	HBS	Total consumption (EUR_HE00)
CIT	HFCS	Business wealth (da1140+da2104+da2105)
Social Contributions	EU-SILC	Labor income (PY010G)
Disability pension	EU-SILC	Disability benefits (PY130G)
Old-age pension	EU-SILC	Old-age benefits (PY100G)
Sickness allowance	EU-SILC	Sickness benefits (PY120G)
Survivor pension	EU-SILC	Survivor benefits (PY110G)
Unemployment subsidy	EU-SILC	Unemployment benefits (PY090G)

Note: Time samples used were the 2019 wave of EU-SILC, the 2015 wave for the HBS, and wave 3 (2017) of the HFCS.

Table C.2: Macrodata Sources.

Aggregate	Variable
<i>Demographic data (from Eurostat)</i>	
Population projections	proj_19n
<i>Fiscal data (from Eurostat - gov_10a_ggfa data-set)</i>	
Personal income	D51A
Property	D29A
Value-added	D211
Corporate income	D51B
SS contributions	D611+D612+D613
Disability pension	GF1001
Sickness allowance	GF1001
Old-age pension	GF1002
Survivor pension	GF1003
Unemployment subsidy	GF1005
Education expenditure	GF09
Health expenditure	GF07
<i>Other macro variables</i>	
GDP	Eurostat: CP_MEUR
Potential GDP	AMECO: OVGDP
Elasticities budget items	From Price, Dang, and Botev (2015)
GDP deflator	Eurostat: PD15_EUR
Net Gov. wealth	Eurostat: gov_10a

D Robustness Tests

D.1 Macroeconomic Assumptions

Our methodology requires specifying paths for the future interest rate, i , and the productivity growth rate, γ . In order for the present discounted value of revenues and expenditures to be a finite number, we must have these parameters such that $\mathcal{D} \in (-1, 1)$. Otherwise, the intertemporal budget constraint loses its meaning, and the exercise becomes inconsistent. Within these bounds, we perform a sensitivity analysis under two extreme scenarios: the first, with the demographic profiles not growing in real terms; and the second, the knife-edge case in which the profiles' growth rate is nearly equal to the interest rate.²⁶ Table D.1 shows the rebalancing tax increase for these two different scenarios. θ_τ changes very little with different macroeconomic assumptions.

Take the equation that implicitly defines θ_τ , Equation (6), and divide by the discount factor. We get an expression where the discount factor only appears in the second term of the left-hand side of the equation. That term corresponds to the stock of public debt, which is significantly smaller than the sum of the deficits generated by population aging. Hence, the effects on θ_τ are very small.

²⁶In practice, we set $i = \gamma - \varepsilon$, with $\varepsilon = 10^{-7}$.

Table D.1: Imbalance Metrics under Different Macroeconomic Assumptions.

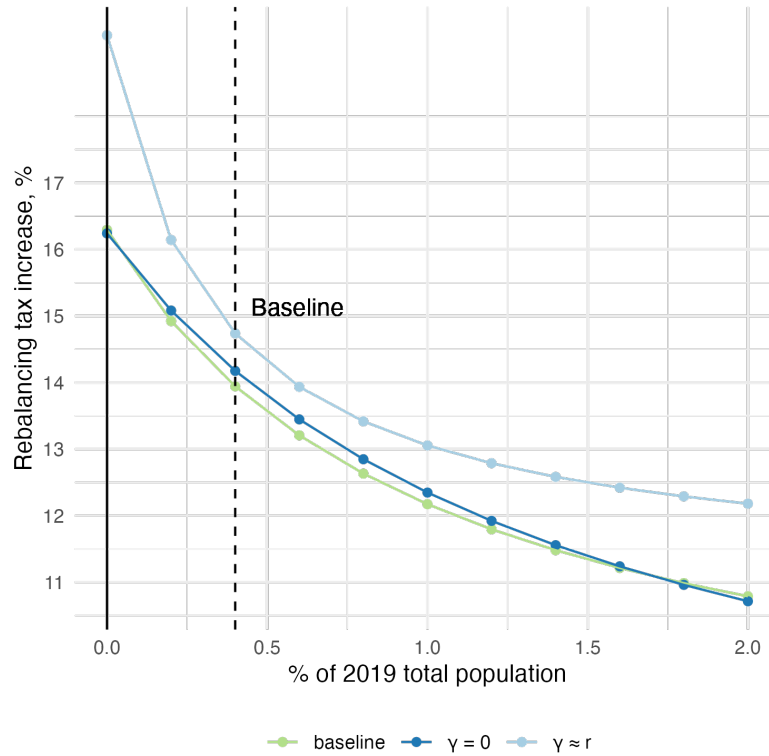
Macro Scenario	θ_τ
Baseline	14.0%
$\gamma = 0$	14.2%
$\gamma = r$	14.9%

Note: The value of θ_τ reported corresponds to the weighted average of the rebalancing tax increase of each country computed according to Equation (6), weighted by the potential GDP of 2019 of each EA country.

Source: Authors' calculations.

To show that macroeconomic hypotheses do not play a role in the rising costs to public finances of reducing migration, we also recompute the frontier between the rebalancing tax increase and the immigration level. It follows from the fact that $\mathcal{D}$ has a minimal influence on θ_τ that it also does not affect the convex relationship between migration and the tax adjustment. In Figure D.1, we plot the frontier between the immigration level and θ_τ for the different assumptions regarding the discount factor.

Figure D.1: Frontier between the Level of Net Migration and the Imbalance Factor Implied for Different Values of the Discount Factor, D .



Note: The figure shows the weighted average θ_τ across countries using the 2019 potential GDP as weights, for the different net migration scenarios. θ_τ is computed according to Equation (6), under different values for the discount factor, $\mathcal{D}$. The dashed line is the baseline net migration value.

Source: Authors' calculations.

D.2 Immigrants Offspring Fertility

In the baseline scenario, we assume that the fertility of immigrants' offspring equals that of the native population. In this robustness exercise, we show that the convexity result is independent of this assumption. For that, we consider the case in which the descendants of immigrants retain the same fertility parameters as their parents. In Table D.2 we report the imbalance metric for the alternative offspring fertility assumption.

Table D.2: Imbalance Metrics under Different Assumptions for Immigrants' Offspring Fertility.

Offspring Fertility Scenario	θ_τ
Native (baseline)	14.0%
1st generation	14.1%

Note: The value of θ_τ reported corresponds to the weighted average of the rebalancing tax increase of each country computed according to Equation (6), weighted by the potential GDP of 2019 of each EA country.

Source: Authors' calculations.

The difference in the rebalancing tax increase between the two scenarios is 0.1 percentage points. This illustrates that our baseline results are robust to the hypothesis that immigrants' offspring have a higher fertility rate. Figure D.2 shows the frontier between the level of immigration and the rebalancing tax increase. The convex relationship between the two remains under the alternative assumption.

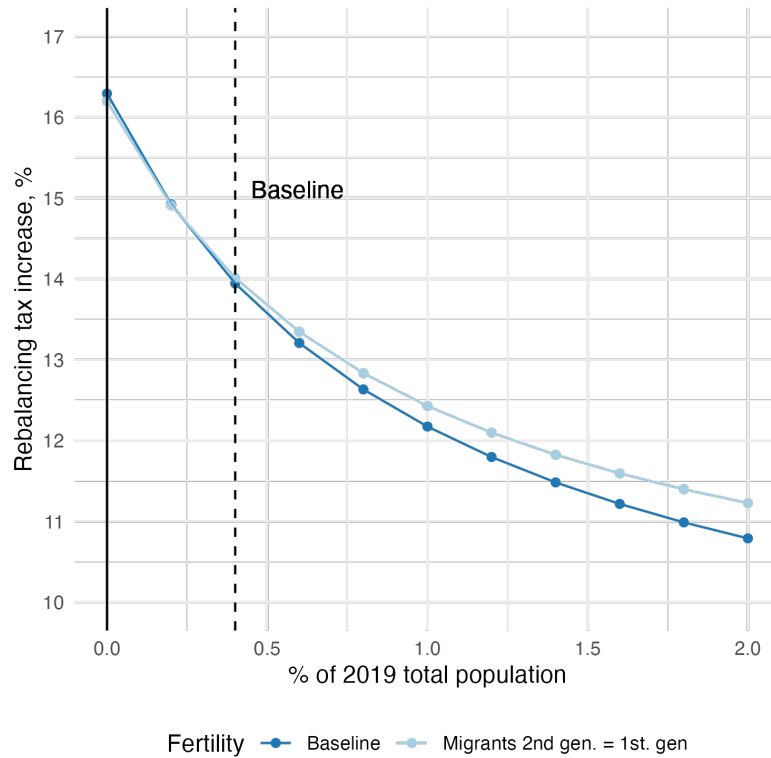
D.3 Immigrants Fertility

Generally, fertility rates are higher in developing countries. It is well known that this translates into higher fertility among migrants. We also observe this demographic behavior in our data. The total fertility rate of residents born in non-EU countries is almost double that of the native population. We perform sensitivity analyses to assess the extent to which migration's impact depends on the fertility of non-EU immigrants. We consider two cases:

1. High Fertility: Constant to the Base Year. This amounts to keeping the fertility of non-EU-born immigrants entering the euro area in 2019 constant. This is an optimistic scenario, as it seems more plausible that fertility will decline over time: the data on fertility rates of developing countries show they have been decreasing in recent decades.

2. Low Fertility: Convergent to the Natives' Value. The second alternative scenario is rather pessimistic. We set the fertility rate of non-EU immigrants to converge towards

Figure D.2: Frontier between the Level of Immigration and the Imbalance Factor under Different Assumptions for Immigrants' Offspring Fertility.



Note: The figure shows the weighted average θ_τ across countries using the 2019 potential GDP as weights, for the different net migration scenarios. θ_τ is computed according to Equation (6), scenarios of the non-EU born fertility rates. The dashed line is the baseline net migration value.

Source: Authors' calculations.

1.6 children per woman, the same as that of nationals, by 2100.

These scenarios are illustrated in the time path for the total fertility rate, shown in Figure B.1.

Table D.3 shows the values of the rebalancing tax increase for the different fertility scenarios. The differences are quite small across fertility scenarios.

Table D.3: Imbalance Metrics under Different Non-EU Immigrant Fertility Rates.

Immigrants Fertility Scenario	θ_τ
Baseline	14.0%
High fertility	13.6%
Low fertility	14.2%

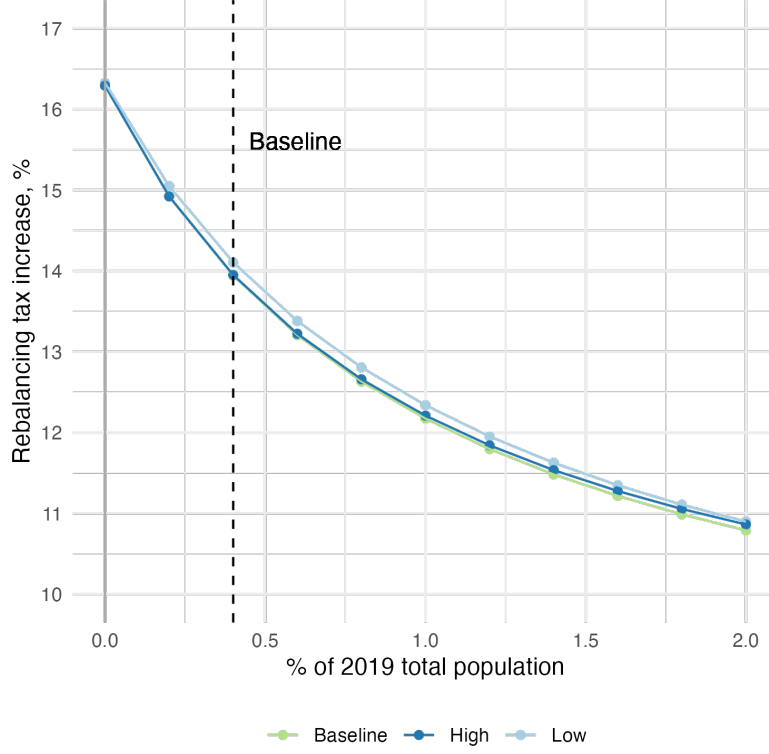
Note: The value of θ_τ reported corresponds to the weighted average of the rebalancing tax increase of each country computed according to Equation (6), weighted by the potential GDP of 2019 of each EA country.

Source: Authors' calculations.

The convex relationship between the migration restriction and the rebalancing tax increase is robust to two fertility scenarios. Figure D.3 plots the frontier for the two

immigrants' fertility hypotheses.

Figure D.3: Frontier between the Level of Immigration and the Imbalance Factor for Different Immigrants' Fertility Hypotheses.



Note: The figure shows the weighted average θ_τ across countries using the 2019 potential GDP as weights, for the different net migration scenarios. θ_τ is computed according to Equation (6), scenarios of the non-EU born fertility rates. The dashed line is the baseline net migration value.

Source: Authors' calculations.

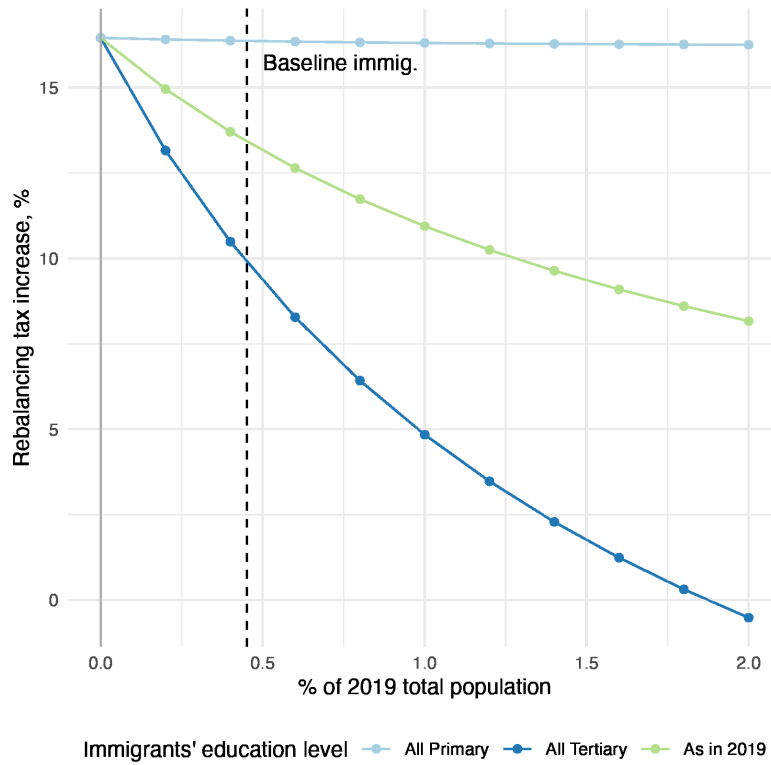
D.4 Immigrants Education Composition

In the baseline scenario, we assume that the educational composition of new immigrants remains constant at the 2019 level. In this appendix, we show that this hypothesis does not alter our key result: the rebalancing tax increase exhibits decreasing returns to immigration.

Figure D.4 shows the frontier between the rebalancing tax increase and net migration for alternative, rather extreme, assumptions for the education composition of migration. The green line represents the baseline, as in the main text; the light blue line represents the frontier if immigration consists only of individuals with primary education; and the dark blue line represents the frontier when immigration consists only of individuals with higher (university) education.

In all three cases, the frontier is negatively convex, as described above. Recall that

Figure D.4: Frontier between the Level of Immigration and the Imbalance Factor for Different Immigrants' Education Composition Hypotheses.



Note: The figure shows the weighted average θ_τ across countries using the 2019 potential GDP as weights, for the different net migration scenarios. θ_τ is computed according to Equation (6) for different hypotheses about the education composition of immigrants. The dashed line is the baseline net migration value.

Source: Authors' calculations.

this means immigration lowers the rebalancing tax increase but at a decreasing rate. However, convexity varies with the educational composition of immigration. In the case where immigration consists only of a low-educated population, the frontier is almost flat, indicating that more immigration does not significantly affect θ_τ . On the other hand, if immigration consists only of highly educated individuals, the frontier between the size of the immigration inflow and the increase in the rebalancing tax is steeper, with immigration bringing more positive effects for public finances.

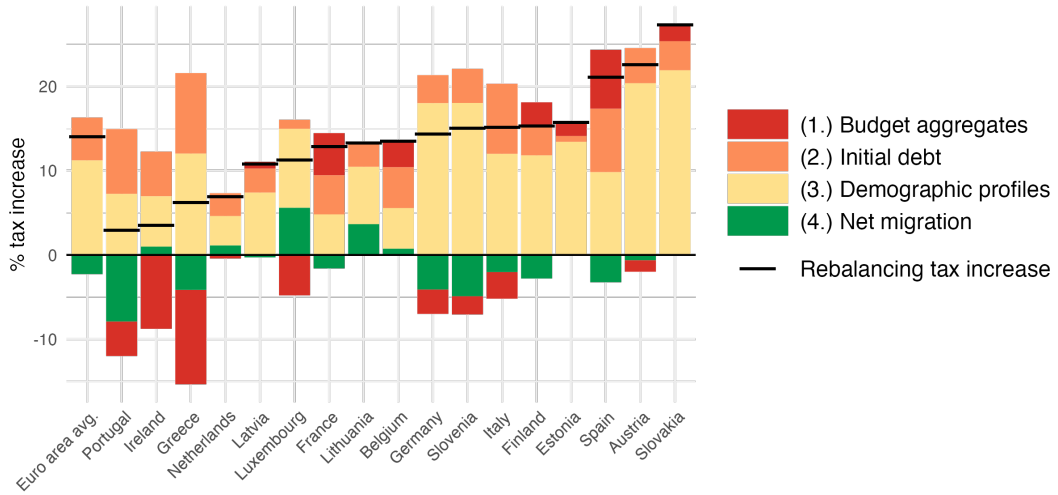
Overall, these results highlight that even with lower education levels than those of the resident population, non-EU immigrant flows can help alleviate the fiscal burden of aging. The fiscal costs of building walls would be significant and rising, even if the education levels of non-EU immigrants were lower than they are now. This also indicates that restricting high-skilled migration is more damaging for the long-run sustainability of public finances than low-skilled migration.

E Additional Figures

E.1 Country-specific Rebalancing Tax Increase Decomposition

To understand what is behind the observed differences across countries, we develop a decomposition of the rebalancing tax increase that allows us to calculate the contributions of the initial debt, the initial aggregate fiscal imbalance, the demographic structure of the budget, and migration. For that, we recompute the rebalancing tax increase, holding different components fixed. This yields the contribution of different factors to the metric. Figure E.1 shows the results of this decomposition for each country and for the EA average. We proceed in four steps, leading up to the baseline θ_τ .

Figure E.1: Decomposition of the Rebalancing Tax Increase for EA Countries.



Note: The figure reports the rebalancing tax increase, θ_τ , for the different countries of the EA and the EA average (solid black line). It also shows the decomposition of this metric in four components: initial debt, budget aggregates, demographic profiles, and net migration.

Source: Authors' calculations.

Recall the expression of the rebalancing tax increase

$$\sum_{s=0}^{\infty} \sum_i \sum_{x \in X} D^s \left[g_{t,x}^i - (1 + \theta_\tau) \tau_{t,x}^i \right] P_{t+s,x}(f, m, M) + B_{t-1} = 0,$$

where we make explicit that population is a function of fertility, f , mortality, m , and net migration, M . We compute successive values of the rebalancing tax increase under different assumptions about macro aggregates, profiles, and population, to isolate the contributions of the initial public debt, the initial primary balance, the demographic profiles, and net migration.

Define $\tau_{\bar{t}} \equiv \sum_i \sum_x \tau_{\bar{t},x}^i$, corresponding to the per capital revenue in period $\bar{t}$, and likewise for $g_{\bar{t}}$. Then, we obtain the following measures.

1. Contribution of Initial Debt. Set $g_{\bar{t}}^* = \tau_{\bar{t}}$, such that the budget is balanced.

$$\sum_{s=0}^{\infty} \sum_{x \in X} D^s \left[g_{\bar{t}}^* - (1 + \theta^{debt}) \tau_{\bar{t}} \right] P_{\bar{t}+s,x}(f, m, 0) + B = 0$$

We start with uniform demographic profiles—all individuals pay the same net taxes—, a balanced initial primary budget, and net migration equal to zero. The only source of imbalances in the IGBC—Equation (5)—is the initial debt, as the revenue collected matches the expenditure done in the same year. The rebalancing tax increase, in this case, captures only the contribution of the initial debt stock, which is always positive—it is the permanent tax increase necessary to sustain the public debt, even if the budget were balanced.

2. Contribution of the Initial Deficit.

$$\sum_{s=0}^{\infty} \sum_{x \in X} D^s \left[g_{\bar{t}} - (1 + \theta^{debt} + \theta^{fiscal}) \tau_{\bar{t}} \right] P_{\bar{t}+s,x}(f, m, 0) + B = 0$$

Second, we set the budget aggregates to the observed base-year values while keeping uniform demographic profiles. The rebalancing tax increase, in this case, covers the initial public debt stock and the initial primary balance repeated over time. By contrast with the previous, we obtain the contribution of the budget aggregates, which is positive for some countries and negative for others, depending on whether the primary balance in 2019 is a deficit or a surplus.

3. Contribution of the Demographic Profiles.

$$\sum_{s=0}^{\infty} \sum_i \sum_{x \in X} D^s \left[g_{\bar{t},x}^i - (1 + \theta^{debt} + \theta^{fiscal} + \theta^{dem}) \tau_{\bar{t},x}^i \right] P_{\bar{t}+s,x}(f, m, 0) + B = 0$$

Third, we reintroduce the demographic profiles from the data, where different age or education groups pay different net taxes. Now, changes in the population's age structure affect the budget balance over time. The change in the θ_{τ} metric from this step gives the impact of demographic change on the rebalancing tax increase. For all countries, the impact on θ_{τ} is always positive: the current structure of government budgets in Europe is not prepared for an aging society.

4. Contribution of the Net Migration Flow.

$$\sum_{s=0}^{\infty} \sum_i \sum_{x \in X} D^s \left[g_{t,x}^i - (1 + \theta^{debt} + \theta^{fiscal} + \theta^{dem} + \theta^M) \tau_{t,x}^i \right] P_{t+s,x}(f, m, M) + B = 0$$

Finally, we add the net migration flow, which changes population dynamics. This last step recovers the baseline θ_τ .

Analyzing the decomposition of the rebalancing tax increase in the EA, the budget-aggregate contribution has a negligible impact on θ_τ , given the nearly balanced primary deficit recorded in 2019. The initial public debt stock contributes 5.1 percentage points to the rebalancing tax, while the demographic composition of the budget accounts for 11.2 percentage points. Additionally, net migration has a positive effect, reducing θ_τ by 2.3 percentage points, as shown before.

The Figure enables cross-country comparisons of each factor's contribution. For instance, France and Germany have very similar overall rebalancing tax increases; however, their root causes differ. The impact of demographic change is more favorable in France, but the initial budget balance position is much worse, contributing almost one-third of the value of θ_τ . Germany, on the other hand, has a favorable initial fiscal position but an ill-prepared budget demographic structure, given the aging trends it will face.

E.2 Expected Contribution at Age 30 for Natives and Migrants

Table E.1: Expected Yearly Lifetime Contributions and Tax Increase for Taxpayers Aged 30 in 2019, by Country.

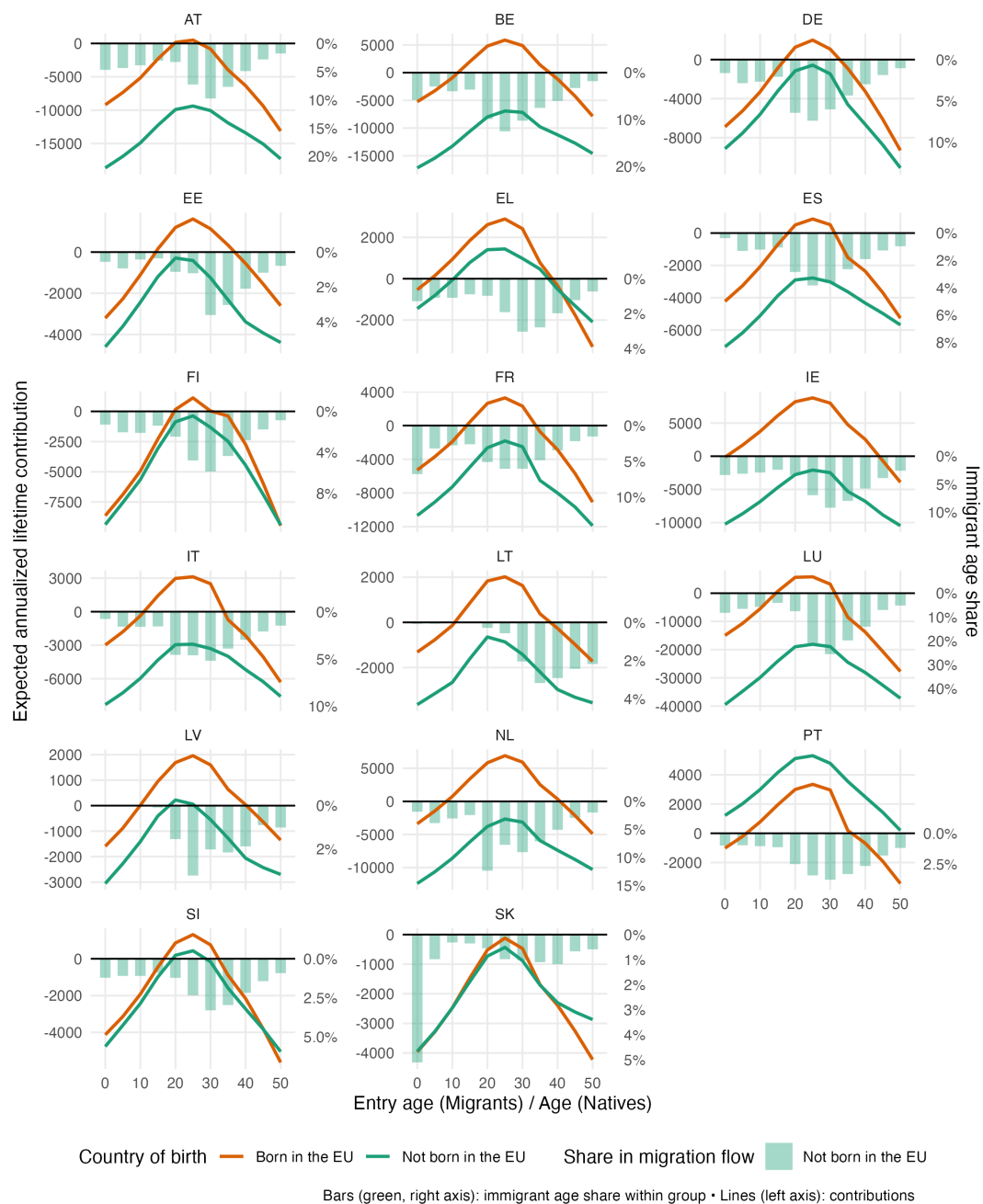
Country	CoB Group	Net Contr.	Tax paym.	Tax increase by θ_τ		Diff.
				baseline	no migration	
AT	EU-born	-825	44619	10067	10343	-276
AT	Non EU-born	-10064	31276	7057	7250	-193
BE	EU-born	4883	45596	6156	5819	337
BE	Non EU-born	-7159	29975	4047	3826	222
DE	EU-born	1073	37037	5312	6828	-1515
DE	Non EU-born	-1462	31978	4587	5895	-1308
EE	EU-born	1129	15412	2428	2423	5
EE	Non EU-born	-1248	13119	2067	2062	4
EL	EU-born	2421	19149	1192	1986	-793
EL	Non EU-born	981	14339	893	1487	-594
ES	EU-born	517	24333	5136	5926	-790
ES	Non EU-born	-3027	16675	3519	4061	-541
FI	EU-born	32	44735	6852	8103	-1251
FI	Non EU-born	-1329	39245	6011	7109	-1098
FR	EU-born	2311	43177	5549	6243	-693
FR	Non EU-born	-2511	35810	4603	5178	-575
IE	EU-born	7997	44885	1573	1131	442
IE	Non EU-born	-2481	32674	1145	823	322
IT	EU-born	2499	32898	4978	5644	-666
IT	Non EU-born	-3300	22482	3402	3857	-455
LT	EU-born	1629	11712	1553	1125	428
LT	Non EU-born	-1417	9009	1195	865	329
LU	EU-born	3213	100285	11282	5676	5605
LU	Non EU-born	-18937	71989	8098	4075	4024
LV	EU-born	1596	11335	1220	1253	-33
LV	Non EU-born	-548	9322	1004	1031	-27
NL	EU-born	5931	43917	3033	2540	493
NL	Non EU-born	-3127	35649	2462	2062	400
PT	EU-born	2962	21489	631	2327	-1696
PT	Non EU-born	4785	20155	592	2183	-1591
SI	EU-born	754	19987	3003	3981	-978
SI	Non EU-born	-161	16856	2533	3358	-825
SK	EU-born	-470	12742	3478	3488	-10
SK	Non EU-born	-881	9958	2718	2726	-8
EA	EU-born	2,160	37,123	5,203	6,048	-845
EA	Non EU-born	-2,697	29,369	4,116	4,785	-669

Note: Annuity values, in Euros per year. The table shows, for taxpayers aged 30 in 2019 in each country, by country of birth group, the average projected lifetime net contribution and tax payments, and, for the baseline and no-migration scenario, the additional payments corresponding to the rebalancing tax increase, along with their difference.

Source: Authors' calculations.

E.3 Expected Contribution at Different Ages and Age Structure of Migration Flow

Figure E.2: Expected Annualized Lifetime Contribution by Age and Country of Birth, and Age Structure of Immigration, for the Different EA Countries.

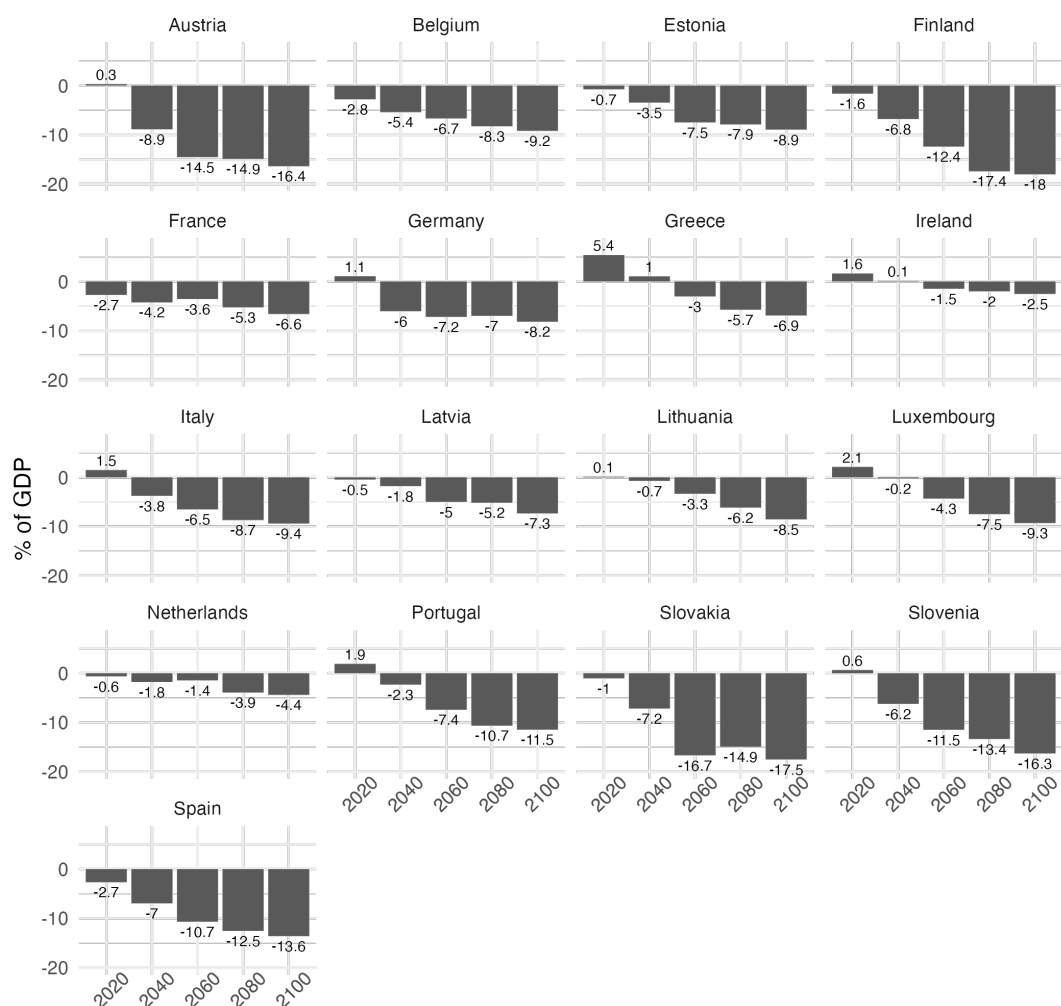


Note: Annuity values, in Euros per year.

Source: Authors' calculations.

E.4 Country-specific Primary Balance

Figure E.3: Counterfactual Primary Balance Implied by the Population Projections for the Different EA Countries.

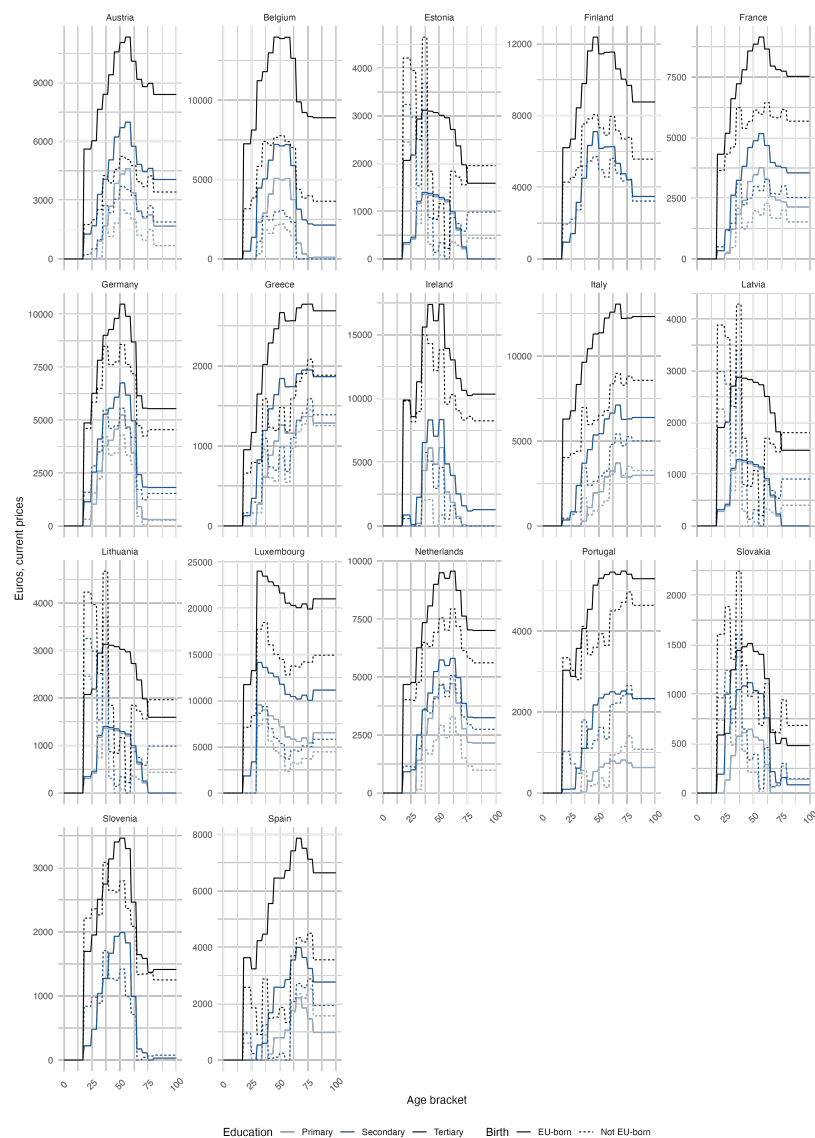


Note: The Figure shows the country-specific primary balance projection for selected years in percentage of the potential GDP.

Source: Authors' calculations.

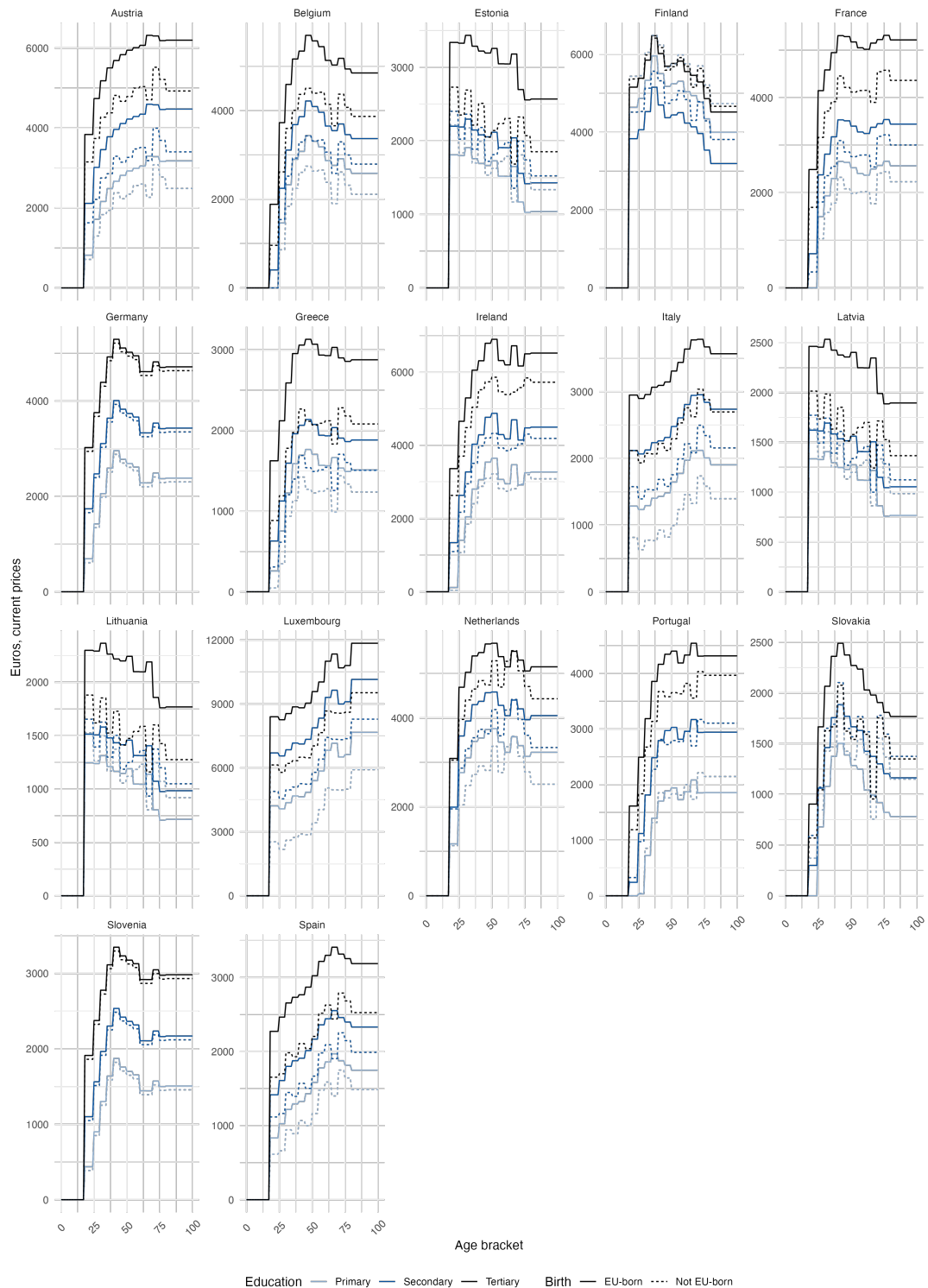
E.5 Country-specific Estimations

Figure E.4: Demographic Profiles of Income Tax Payments.



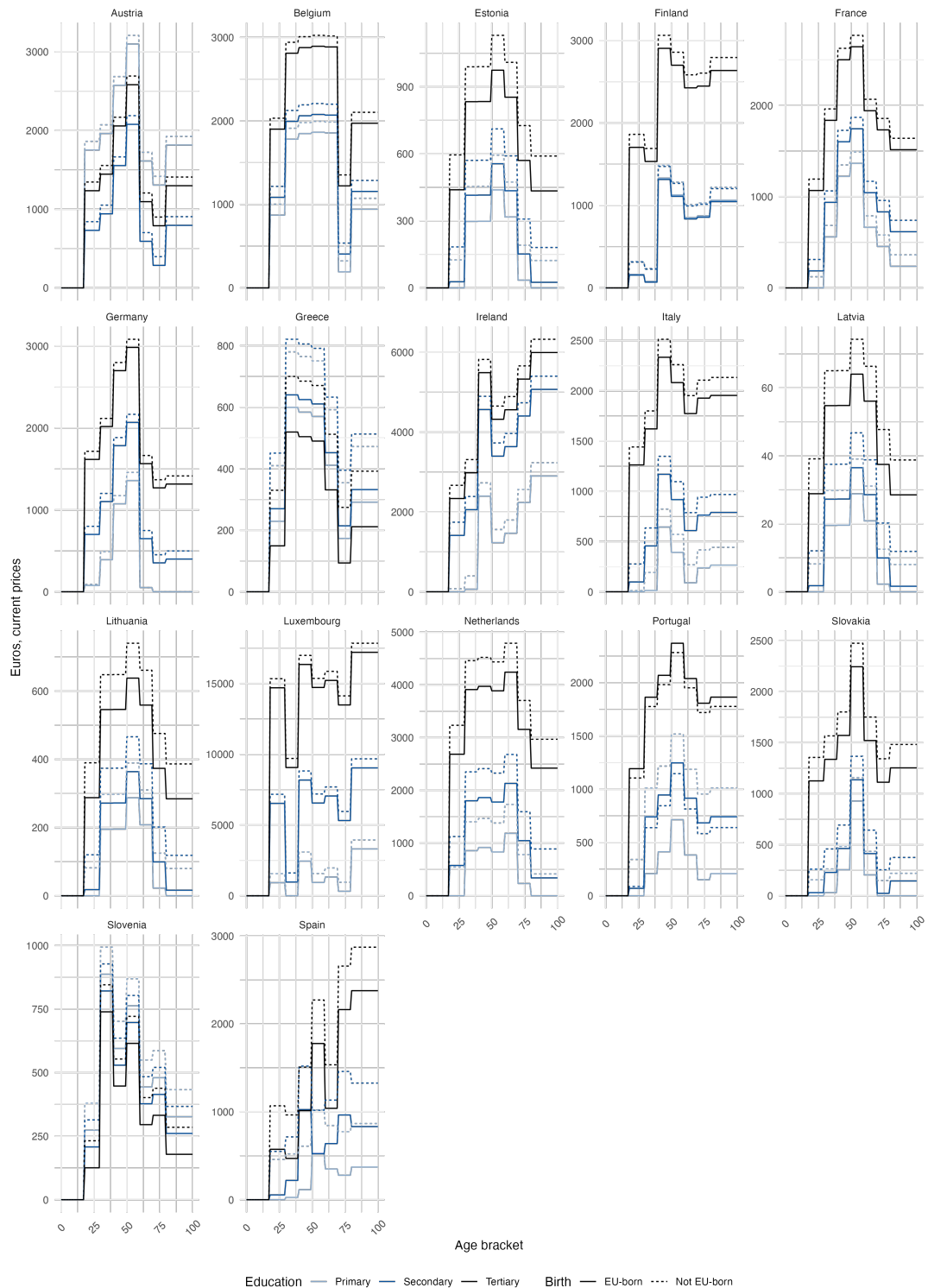
Note: Income tax is assigned to household members in proportion to their respective share in the total household income. Data is plotted for each country, age bracket, education level, and country of birth.
Source: Authors' calculations based on the EU-SILC 2019 wave and Eurostat.

Figure E.5: Demographic Profiles of Consumption.



Note: Total consumption assigned to household members in proportion to their respective share in the total household income. Data is plotted for each country, age bracket, education level, and country of birth.
Source: Authors' calculations based on the EU-HBS 2015 wave and Eurostat.

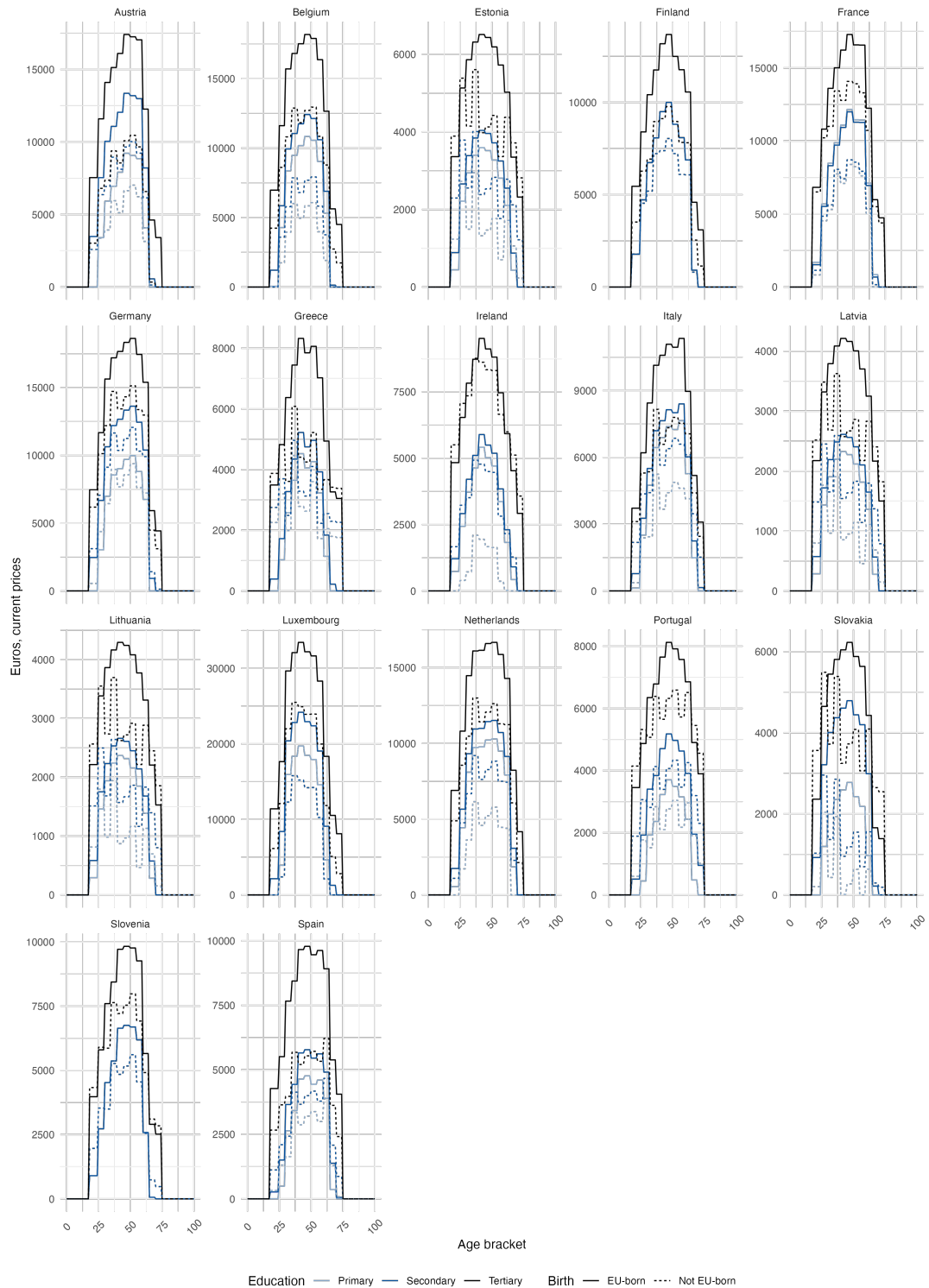
Figure E.6: Demographic Profiles of Business Wealth.



Note: Business wealth holdings are equally split between adult household members. Data is plotted for each country, age bracket, education level, and country of birth.

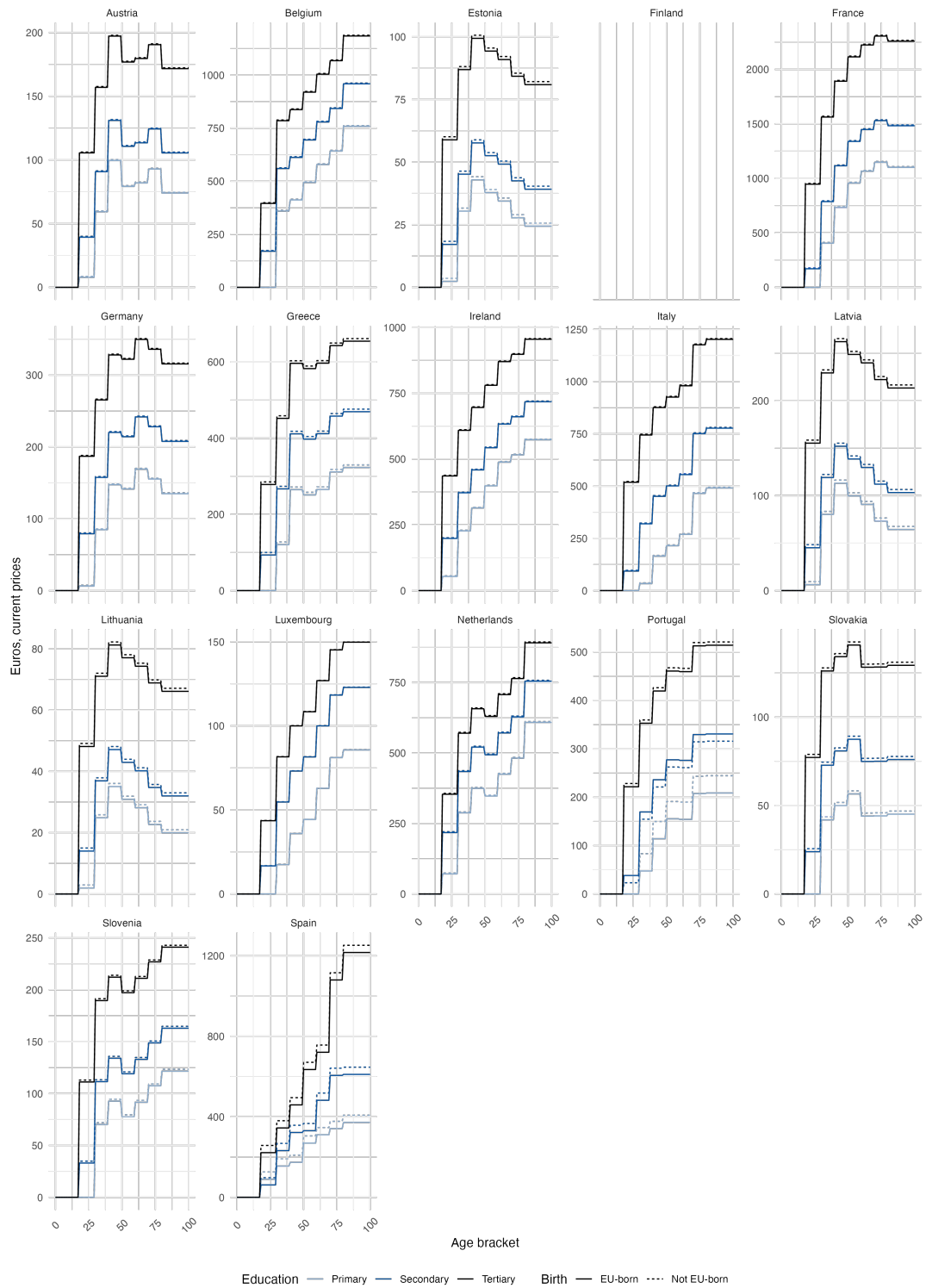
Source: Authors' calculations based on the ECB-HFCS 2017 wave and Eurostat.

Figure E.7: Demographic Profiles of Labor Income.



Note: Data is plotted for each country, age bracket, education level, and country of birth.
Source: Authors' calculations based on the EU-SILC 2019 wave and Eurostat.

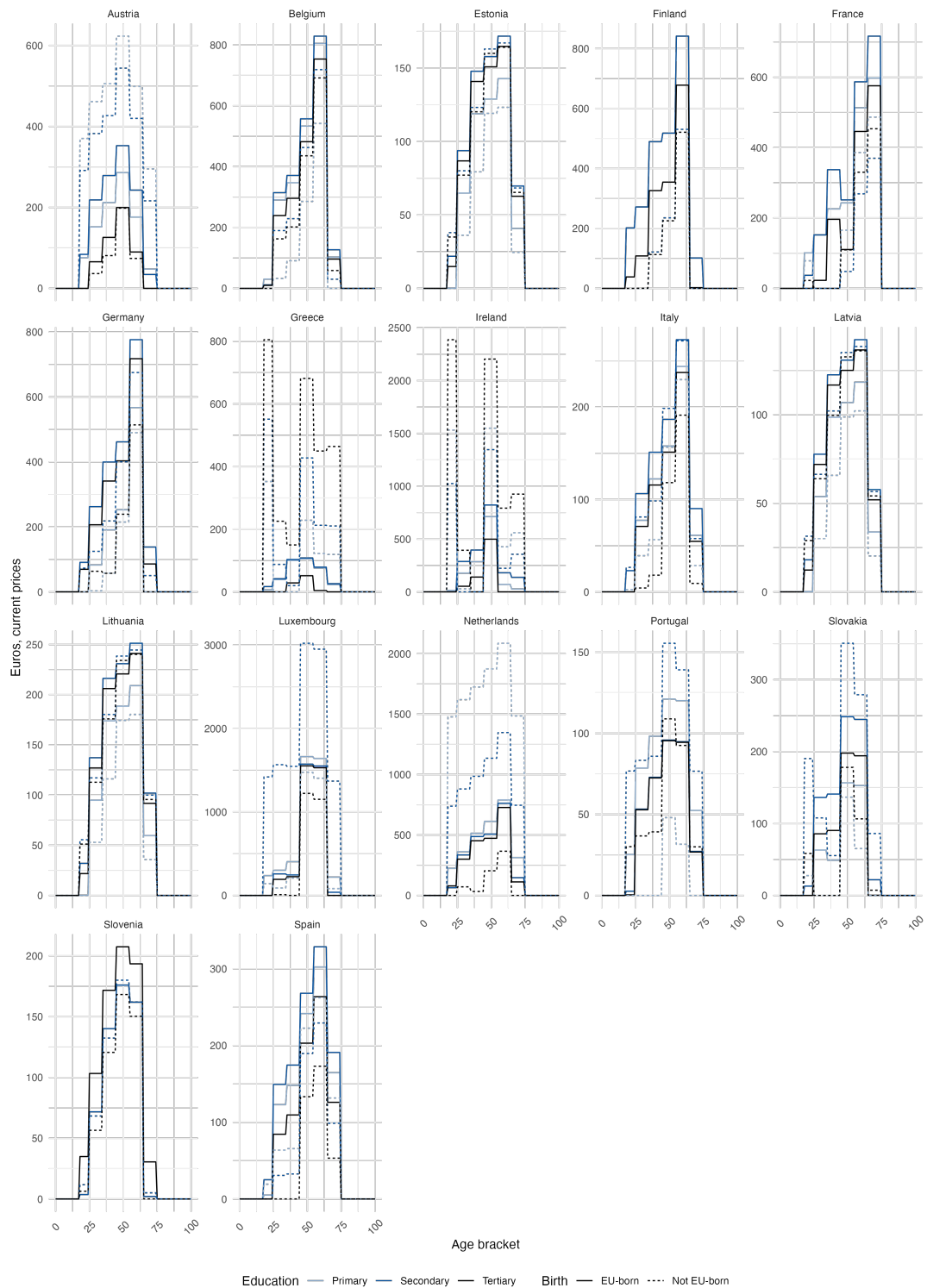
Figure E.8: Demographic Profiles of Wealth Taxes.



Note: Real estate holdings are equally split between adult household members. Data is plotted for each country, age bracket, education level, and country of birth.

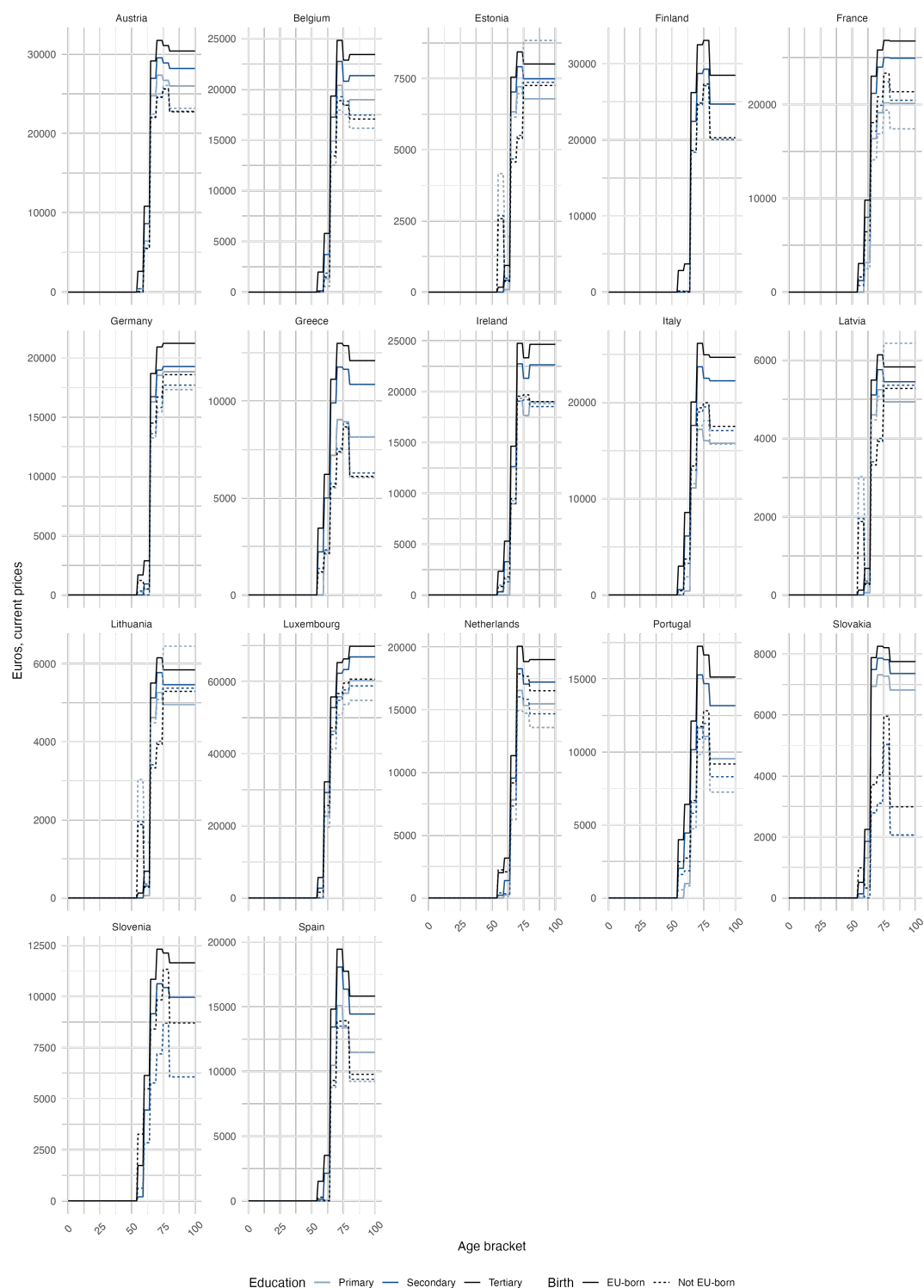
Source: Authors' calculations based on the ECB-HFCS 2017 wave and Eurostat.

Figure E.9: Age Profiles of Sickness Allowance.



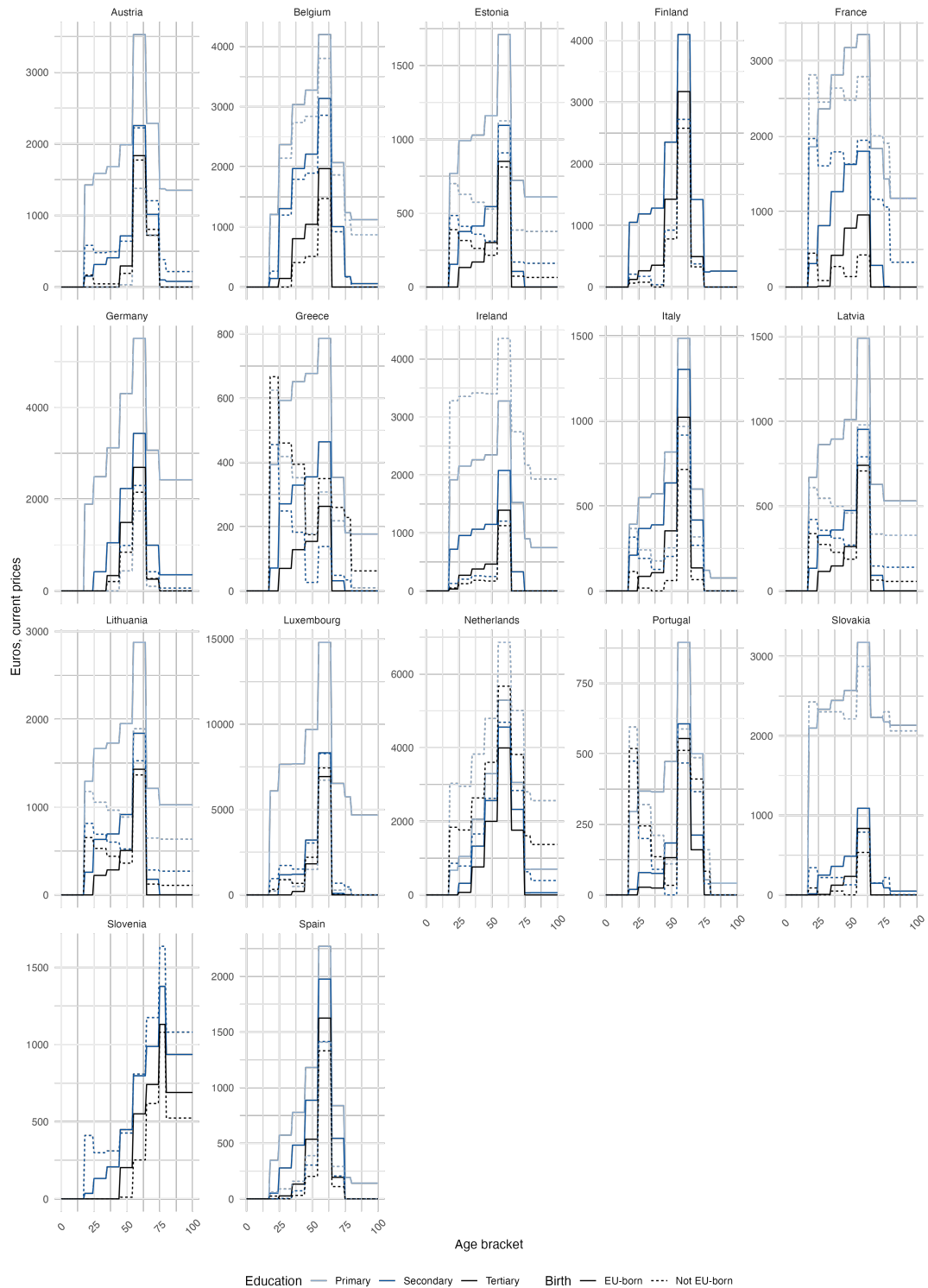
Note: Data is plotted for each country, age bracket, education level, and country of birth. *Source: Authors' calculations based on the EU-SILC 2019 wave and Eurostat.*

Figure E.10: Demographic Profiles of Old-Age Pension.



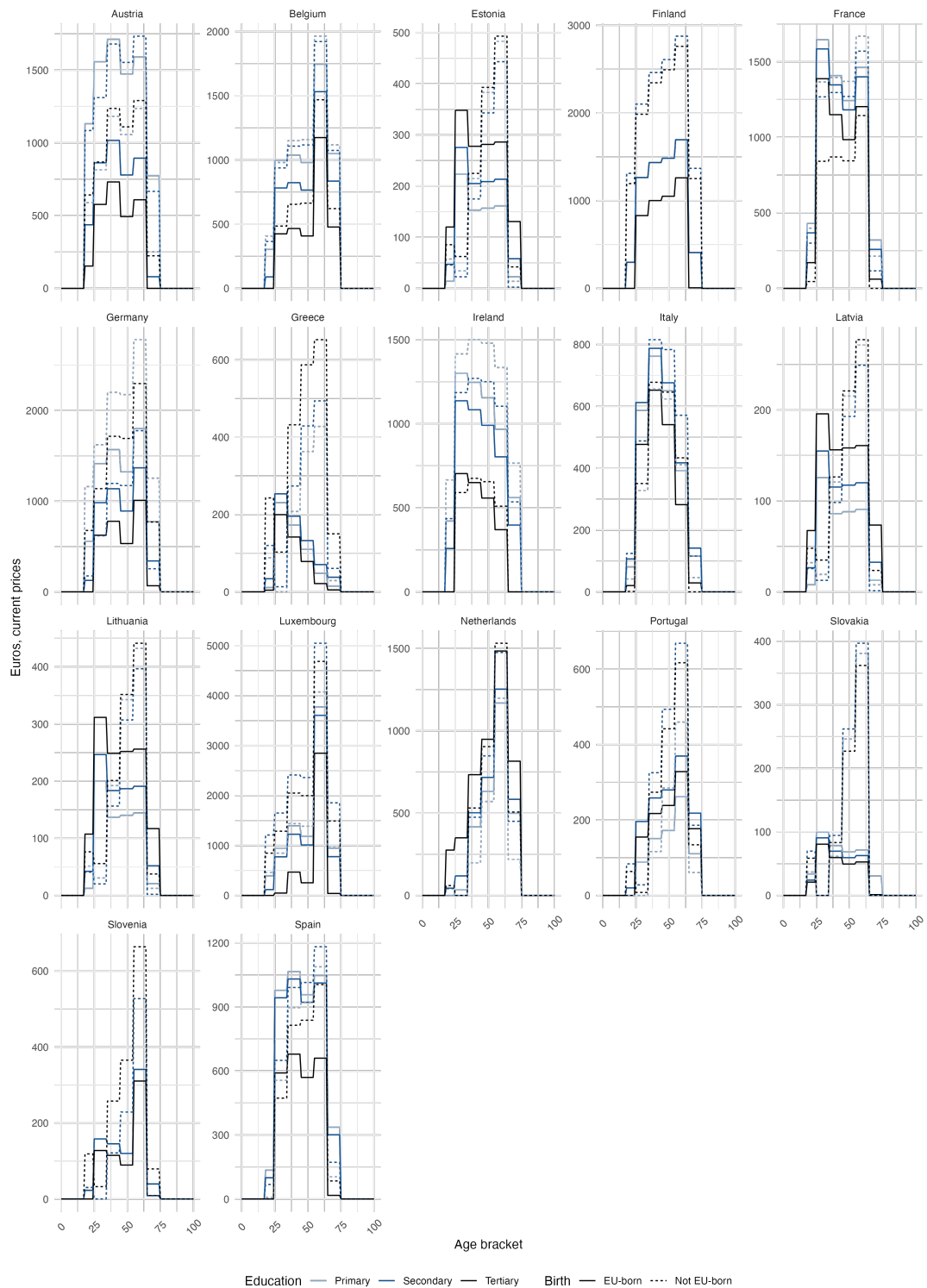
Note: Data is plotted for each country, age bracket, education level, and country of birth.
Source: Authors' calculations based on the EU-SILC 2019 wave and Eurostat.

Figure E.11: Demographic Profiles of Disability Benefits.



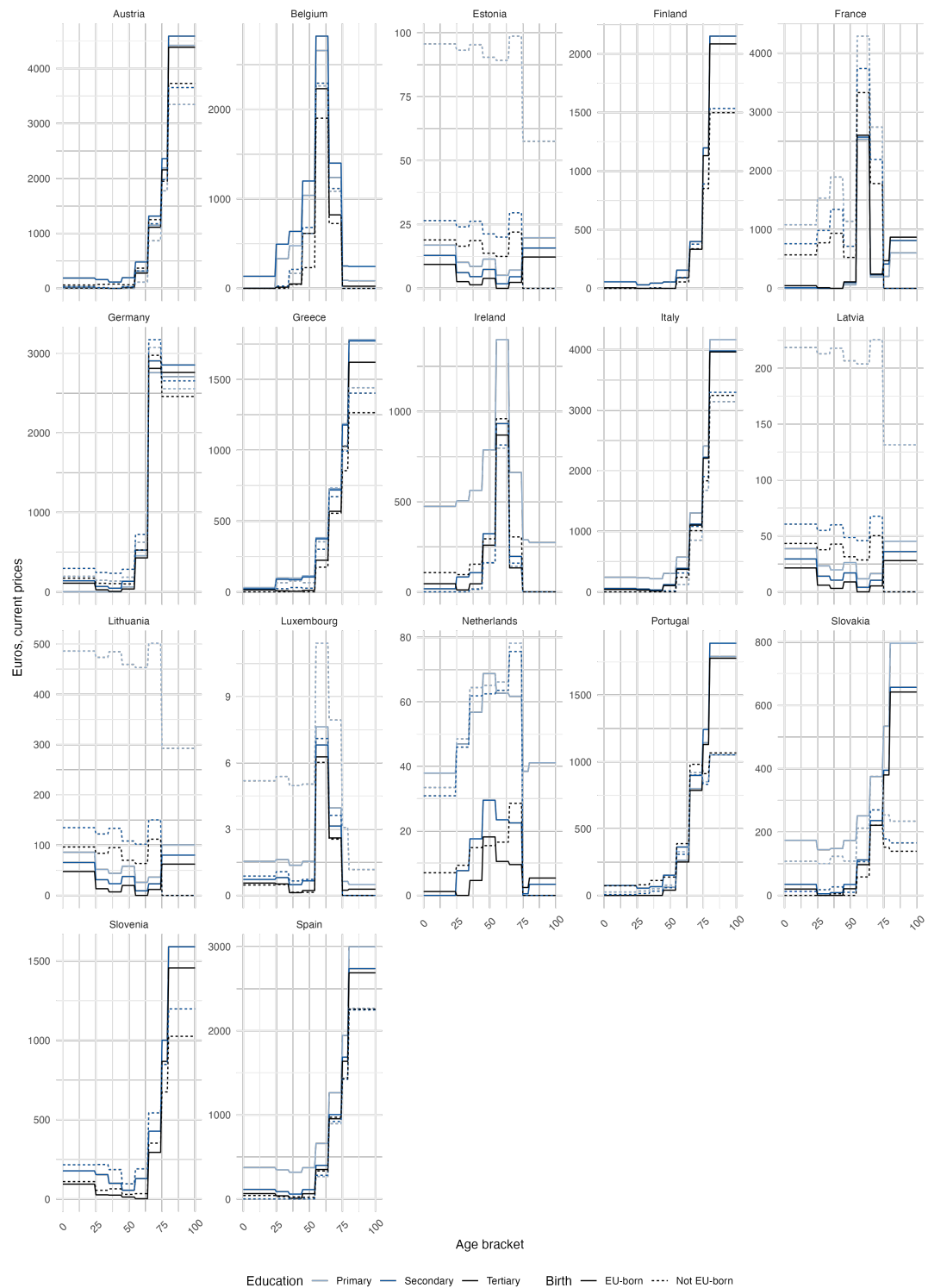
Note: Data is plotted for each country, age bracket, education level, and country of birth.
Source: Authors' calculations based on the EU-SILC 2019 wave and Eurostat.

Figure E.12: Age Profiles of Unemployment Benefits.



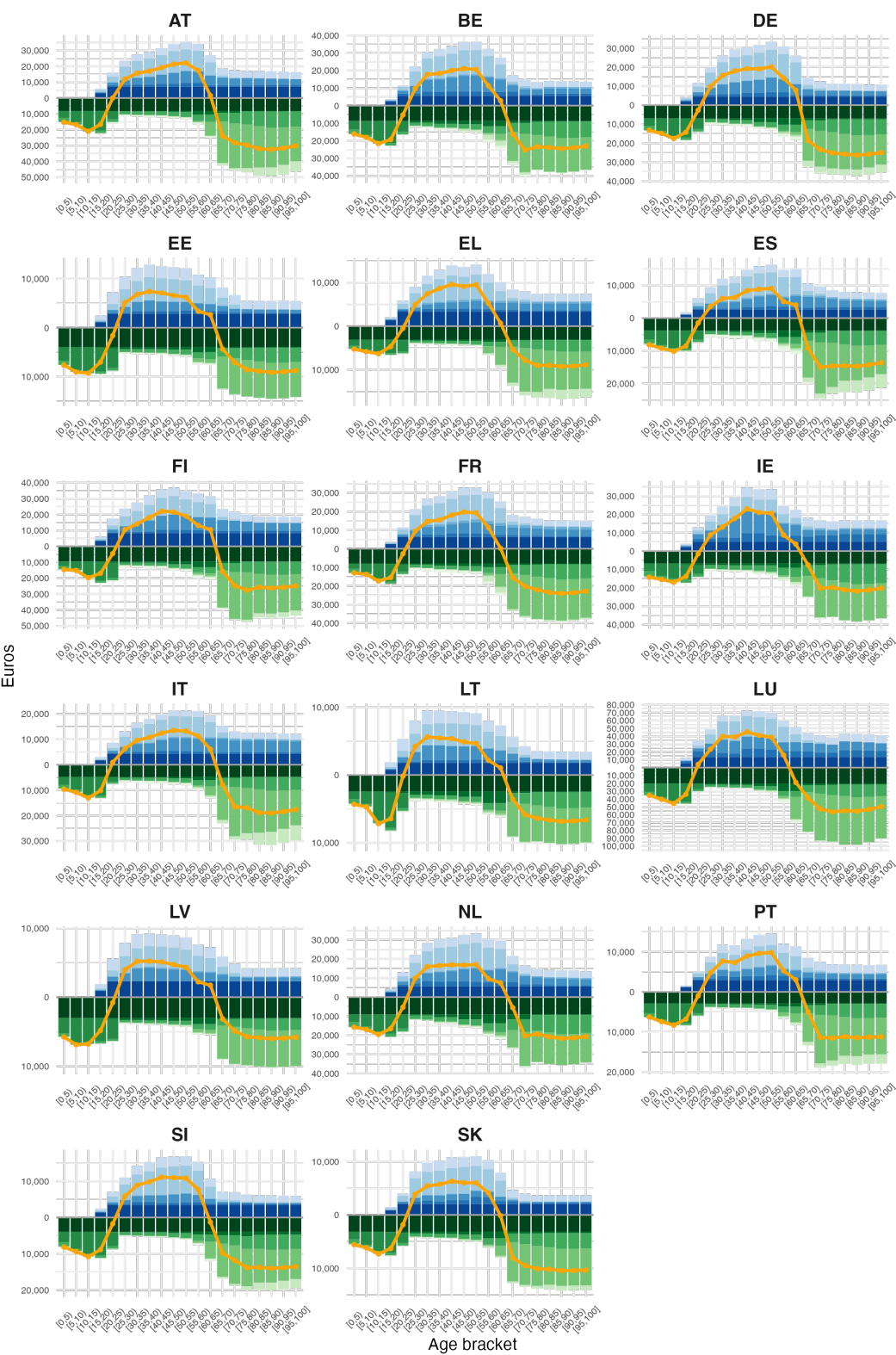
Note: Data is plotted for each country, age bracket, education level, and country of birth.
Source: Authors' calculations based on the EU-SILC 2019 wave and Eurostat.

Figure E.13: Age Profiles of Survivor Pension.



Note: Data is plotted for each country, age bracket, education level, and country of birth.
Source: Authors' calculations based on the EU-SILC 2019 wave and Eurostat.

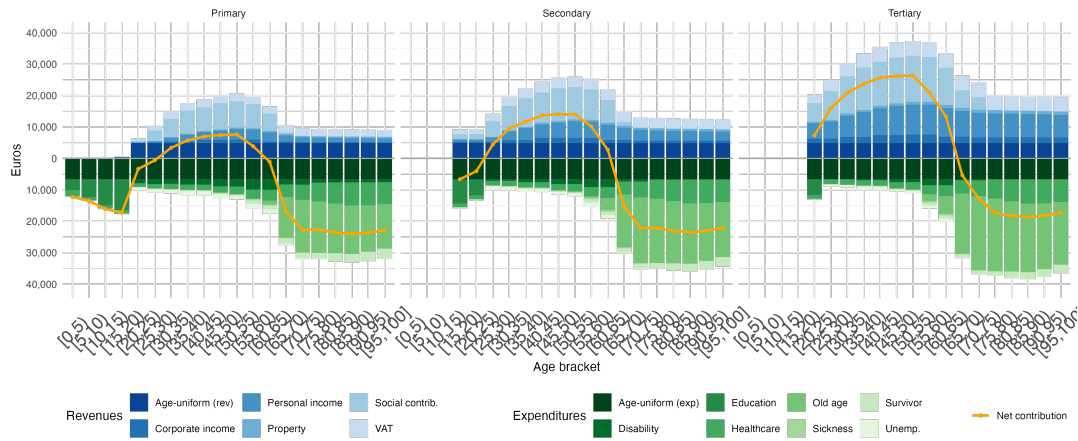
Figure E.14: Demographic Profile of Revenues and Expenditures Per Capita, by Country.



Source: Authors' calculations.

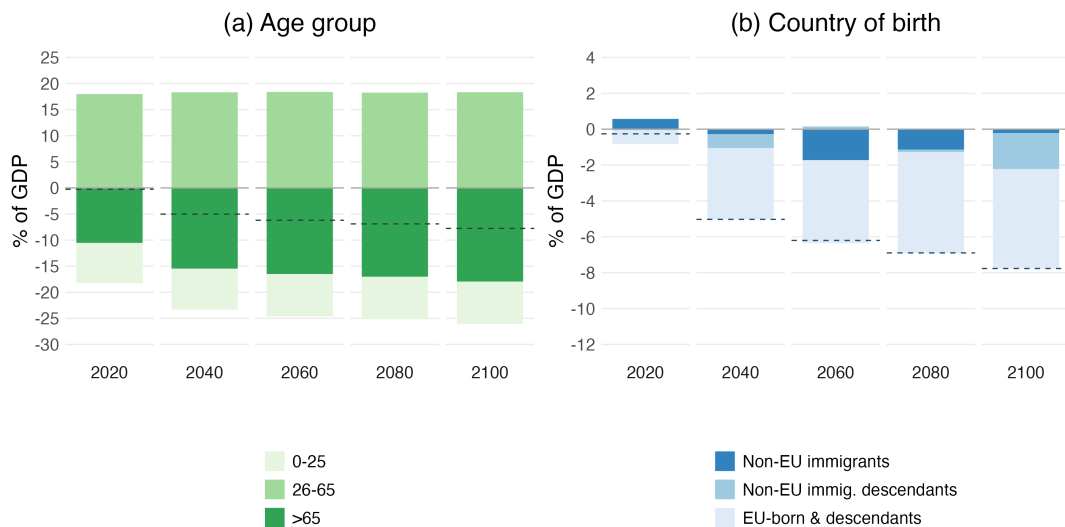
E.6 Other Figures

Figure E.15: Mean Age Profile of Revenues and Expenditures per Capita, by Education Level.



Source: Authors' calculations.

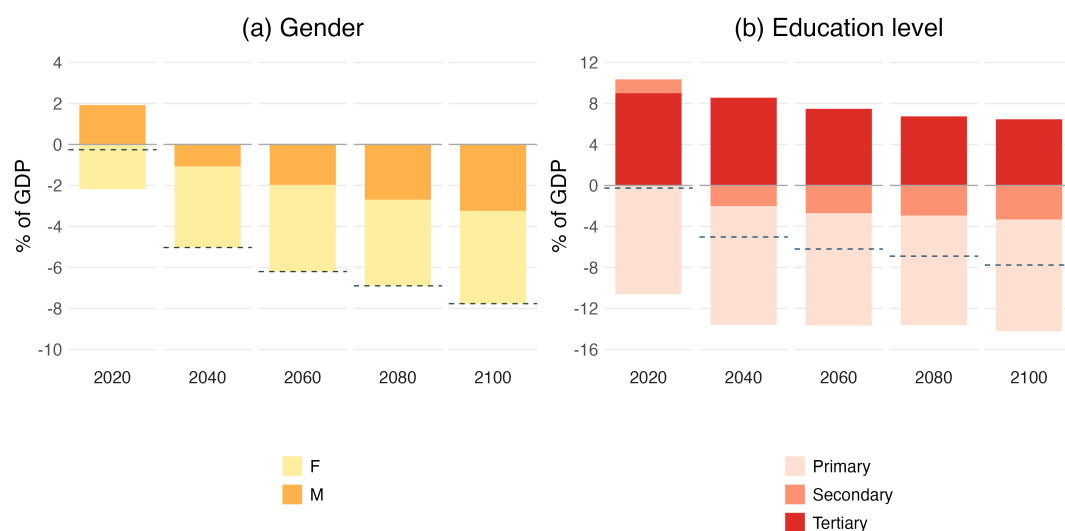
Figure E.16: Counterfactual Primary Balance Implied by the Population Projections Decomposed by Age Group and Country of Birth.



Note: The figure shows the net contribution to the EA primary balance for selected years of the projection of each age group—panel (a)—and of each country of birth group—panel (b). The dashed lines in the figure represent the primary balance. All values are shown as a percentage of the potential GDP.

Source: Authors' calculations.

Figure E.17: Counterfactual Primary Balance Implied by the Population Projections Decomposed by Gender and Education Level



Note: The figure shows the net contribution to the EA primary balance for selected years of the projection of each gender—panel (a)—and of each education level group—panel (b). The dashed lines in the figure represent the primary balance. All values are shown as a percentage of the potential GDP.

Source: Authors' calculations.